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CORELESS INDUCTION FURNACE

Abstract

A coreless induction furnace that is automatically adjustable during assembly operations or melting operations. The furnace may include an outer shell, a set of adjustable shunt systems that operably engages with the outer shell, at least one cooling coil that operably engages with the set of adjustable shunt systems, and a power coil that operably engages with the at least one cooling coil and the set of adjustable shunt systems. The furnace is configured to automatically adjust at least one shunt of the set of adjustable shunt systems relative to at least one of the at least one cooling coil and the power coil. The furnace may also include an adjustable head system that operably engages with the outer shell. The adjustable head system is configured to automatically adjust a head of the adjustable head system relative to the at least one cooling coil and the power coil.

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Background/Summary

TECHNICAL FIELD

[0001] This disclosure is generally directed to induction furnaces, specifically coreless induction furnaces that are automatically adjustable.

BACKGROUND ART

[0002] Induction furnaces, either coreless or channel, operate by transferring high and/or extreme amounts heat energy via a high-voltage primary coil that induces a high current with a low voltage. Generally, these types of furnaces are efficient at melting a wide array of metals while maintaining a low level of melt loss during melting operations. In regards to coreless induction furnaces, coreless induction furnaces include at least one power coil that is the primary heating component of coreless induction furnaces. Generally, the power coils provided in these coreless induction furnaces are highly conductive copper tubing and are wound into helical coils before being placed inside of steel casings or outer shells of coreless induction furnaces. As stated earlier, these coreless induction furnaces generate high and/or extreme amounts of heat energy through these power coils at high current values where frequencies may vary between about 50 cycles per second up to about 10,000 cycles per second, which may generate extreme amounts of power that can be applied to these furnaces.

[0003] With such extreme power and heat, metal materials may be provided in a molten state based on the interaction between the magnetic fields and electrical currents generated by the power coils of these coreless induction furnaces. With these magnetic fields and electrical currents, a stirring reaction is produced inside of the coreless induction furnaces to mix alloys inside of these coreless induction furnaces or to distribute temperature evenly throughout these coreless induction furnaces. However, such extreme power and heat generated and applied inside these furnaces during melting processes via these power coils leads to deformation and destruction to components inside of these furnaces, specifically shunts and heads of these furnaces.

[0004] To combat these issues, conventional coreless induction furnaces may include coils that are fluid-cooled (e.g., water-cooled or other fluids of the like) via a recirculating tower or machine. Conventional induction furnaces may also include cooling coils located above or below the power coils for further cooling to reduce heat. However, such cooling of these power coils may fail to alleviate unwanted pressure and stress placed on the shunts and heads of coreless induction furnaces during melting processes. Moreover, such shunts and heads are constantly maintained at a sustained pressure with the power coils of coreless induction furnaces which ultimately leads to warpage and deformation of these shunts and heads.

[0005] Furthermore, when warpage and deformation of these shunts and heads occur due to numerous melting processes occurring inside of coreless induction furnaces, disassembling and reassembling of these coreless induction furnaces is time consuming, labor intensive, and dangerous to operators. Currently, groups of operators must continuously work for long periods of time in order to disassemble and reassemble these coreless induction furnaces without causing further damage to the coreless induction furnaces and injuries to operators due to extreme pressure applied between the power coils and the shunts and heads. Such disassembly and reassembly of these coreless induction furnaces are vastly time consuming, which leads loss in profits from using these coreless induction furnaces due to lack of melting metals and/or mixing of metal alloys.

SUMMARY OF THE INVENTION

[0006] In one aspect, an exemplary embodiment of the present disclosure may provide an adjustable shunt system of an induction furnace. The adjustable shunt system includes a shunt that operably engages with at least one cooling coil of the induction furnace and a power coil of the induction furnace. The adjustable shunt system also includes a support column that operably engages with an outer shell of the induction furnace. The adjustable shunt system also includes at least one shunt drive assembly that operably engages with the support column and the shunt. The at least one shunt drive assembly is configured to automatically adjust the shunt along an axis angled

relative to the support column.

[0007] This exemplary embodiment or another exemplary embodiment may further include that when the shunt is adjusted by the at least one shunt drive assembly, the at least one cooling coil and the power coil are adjusted with the shunt. This exemplary embodiment or another exemplary embodiment may further include that the at least one shunt drive assembly comprises: a first jack having a first end that operably engages with the shunt and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the shunt along the axis angled relative to the support column. This exemplary embodiment or another exemplary embodiment may further include at least another shunt drive assembly operably engaged with the support column and the shunt; wherein the at least another shunt drive assembly is configured to automatically adjust the shunt along a second axis angled relative to the support column and parallel with the axis of the at least one shunt drive assembly. This exemplary embodiment or another exemplary embodiment may further include that each of the at least one shunt drive assembly and the at least another shunt drive assembly adjusts the shunt independently and separately from one another. This exemplary embodiment or another exemplary embodiment may further include that the at least another shunt drive assembly comprises: a second jack having a first end that operably engages with the shunt and a second end opposite to the first end; and a second motor operably engaged with jack; wherein the second motor is configured to linearly move the jack and the shunt along the second axis angled relative to the support column. This exemplary embodiment or another exemplary embodiment may further include that the support column comprises: a top end; a bottom end opposite to the top end; and a passageway defined therebetween; wherein the first motor and the second motor operably engage with the support column inside of the passageway. This exemplary embodiment or another exemplary embodiment may further include a transition plate operably engaged with the shunt, the at least one cooling coil, and the power coil; wherein the transition plate is positioned between the shunt and the at least one cooling coil and the power coil such that the shunt is spaced apart from the at least one cooling coil and the power coil; and wherein the transition plate and shunt are separate and independent of one another. This exemplary embodiment or another exemplary embodiment may further include that the shunt comprises: a first end; a second end opposite to the first end; a first outer surface extending between the first end and the second end; and a first inner surface extending between the first end of the shunt and the second end of the shunt and opposite to the first outer surface; wherein the first outer surface and the first inner surface are substantially planar surfaces. This exemplary embodiment or another exemplary embodiment may further include that the transition plate comprises: a first end; a second end opposite to the first end; a second outer surface extending between the first end of the transition plate and the second end of the transition plate; and a second inner surface extending between the first end of the transition plate and the second end of the transition plate and opposite to the first outer surface; wherein the second outer surface is a substantially planar surface; and wherein the second inner surface is a curvilinear planar surface. This exemplary embodiment or another exemplary embodiment may further include a track operably engaged with the support column; and a sled assembly operably engaged with the track and operably engaged with the shunt at one of a first end of the shunt and a second end of the shunt opposite to the first end; wherein the sled assembly is configured to guide the shunt along the track. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the shunt and the at least one shunt drive assembly; and a controller electrically connected with the at least one load cell and the at least one shunt drive assembly; wherein the controller is configured to automatically adjust the shunt, via the at least one shunt drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined pressures. This exemplary embodiment or another exemplary embodiment may further include that at least one load cell operably engaged with the shunt and the at least one shunt drive assembly; at least another load cell operably engaged with the shunt and the at least another

shunt drive assembly; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one shunt drive assembly, and the at least another shunt drive assembly; wherein the controller is configured to automatically adjust the shunt, via one or both of the at least one shunt drive assembly and the at least another shunt drive assembly, when at least one of the at least one load cell and the at least another load cell measure a pressure that is outside a range of predetermined pressures.

[0008] In another aspect, an exemplary embodiment of the present disclosure may provide a method of automatically adjusting a shunt of a shunt system of an induction furnace. The method includes steps of engaging at least one shunt drive assembly of the shunt system with a support column of the shunt system; engaging the at least one shunt drive assembly with the shunt of the shunt system; engaging at least one cooling coil of the induction furnace and a power coil of the induction furnace with the shunt; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least one shunt drive assembly, relative to the at least one cooling coil and the power coil.

[0009] This exemplary embodiment or another exemplary embodiment may further include a step of advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include a step of withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least one load cell with the at least one shunt drive assembly and the shunt; and measuring pressure at at least one position on the shunt by the at least one load cell. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a predetermined pressure applied against the shunt; and advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a predetermined pressure applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least another drive assembly with the shunt of the shunt system; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least another drive assembly, relative to the at least one cooling coil and the power coil; wherein the at least one shunt drive assembly and the at least another drive assembly separately and independently adjust the shunt.

[0010] In yet another aspect, an exemplary embodiment of the present disclosure may provide an adjustable head system of an induction furnace. The adjustable head system includes a head. The adjustable head system also includes an apron that operably engages with an outer shell of the induction furnace. The adjustable head system also includes at least one head drive assembly that operably engages with the apron. The at least one head drive assembly is configured to automatically adjust the head along an axis angled relative to the apron.

[0011] This exemplary embodiment or another exemplary embodiment may further include that when the head is adjusted by the at least one head drive assembly, at least one cooling coil of the induction furnace and a power coil of the induction furnace are adjusted with the head. This

exemplary embodiment or another exemplary embodiment may further include that the at least one head drive assembly comprises: a first jack having a first end that operably engages with the head and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the head along the axis angled relative to the apron. This exemplary embodiment or another exemplary embodiment may further include at least another head drive assembly operably engaged with the apron and the head; wherein the at least another head drive assembly is configured to automatically adjust the head along a second axis angled relative to the apron and parallel with the axis of the at least one head drive assembly. This exemplary embodiment or another exemplary embodiment may further include that each of the at least one head drive assembly and the at least another head drive assembly adjusts the head independently and separately from one another. This exemplary embodiment or another exemplary embodiment may further include that the at least another head drive assembly comprises: a second jack having a first end that operably engages with the head and a second end opposite to the first end; and a second motor operably engaged with jack; wherein the second motor is configured to linearly move the jack and the head along the second axis angled relative to the apron. This exemplary embodiment or another exemplary embodiment may further include that the apron comprises: a top end; a bottom end opposite to the first end; and a passageway defined between the top end and the bottom end; wherein the first motor and the second motor operably engage with the apron inside of the passageway. This exemplary embodiment or another exemplary embodiment may further include that the first motor and the second motor operably engage with top end of the apron inside of the passageway. This exemplary embodiment or another exemplary embodiment may further include that the head comprises: a top plate operably engaged with the at least one head drive assembly; wherein the top plate is adjustable relative to the apron and at least one cooling coil of the induction furnace. This exemplary embodiment or another exemplary embodiment may further include that the head further comprises: a refractory member operably engaged with the top plate; wherein the refractory member is configured to engage with at least one cooling coil of the induction furnace to maintain a desired pressure on the at least one cooling coil and a power coil of the induction furnace. This exemplary embodiment or another exemplary embodiment may further include that the top plate comprises: a top end; a bottom end operably engaged with the at least one head drive assembly and opposite to the top end; and a retaining arm extending from the bottom end of the top plate; wherein the bottom end and the retaining arm are configured to retain the refractory member with the top plate. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the head and the at least one head drive assembly; and a controller electrically connected with the at least one load cell and the at least one head drive assembly; wherein the controller is configured to automatically adjust the head, via the at least one head drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined pressures. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the head and the at least one head drive assembly; at least another load cell operably engaged with the head and the at least another head drive assembly; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one head drive assembly, and the at least another head drive assembly; wherein the controller is configured to automatically adjust the head, via at least one of the at least one head drive assembly and the at least head another drive assembly, when at least one of the at least one load cell and the at least another load cell measures pressure that is outside a range of predetermined pressures.

[0012] In yet another aspect, an exemplary embodiment of the present disclosure may provide a method of automatically adjusting a head of a head system of an induction furnace. The method includes steps of engaging at least one head drive assembly of the head system with an apron of the head system; engaging the at least one head drive assembly with the head of the head system;

engaging at least one cooling coil of the induction furnace with the head; and automatically adjusting the head of the head system of the induction furnace, via the at least one head drive assembly, relative to the at least one cooling coil.

[0013] This exemplary embodiment or another exemplary embodiment may further include a step of retracting the head towards the apron and towards the at least one cooling coil and a power coil of the induction furnace, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include a step of withdrawing the head away from the at least one cooling coil and a power coil of the induction furnace and towards the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least one load cell with the at least one head drive assembly and the head; and measuring pressure at least one position on the head by the at least one load cell. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and retracting the head towards the apron and towards the at least one cooling coil and a power coil of the induction furnace, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from the at least one cooling coil and a power coil of the induction furnace and towards the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least another head drive assembly with the head of the head system; and automatically adjusting the head of the head system of the induction furnace, via the at least another head drive assembly, relative to the at least one cooling coil and a power coil of the induction furnace; wherein the at least one head drive assembly and the at least another head drive assembly separately and independently adjust the head.

[0014] In yet another aspect, an exemplary embodiment of the present disclosure may provide a method of monitoring pressure applied to a shunt of an induction furnace during a melting process. The method comprises steps of engaging at least one shunt drive assembly of a shunt system with a support column of the shunt system and the shunt of the shunt system; providing at least one load cell with the shunt at a first position and with the at least one shunt drive assembly of the shunt system; connecting a controller of the induction furnace with the at least one load cell; connecting the controller with the at least one shunt drive assembly; and monitoring pressure applied to the shunt of the induction furnace during the melting process by the at least one load cell.

[0015] This exemplary embodiment or another exemplary embodiment may further include a step of automatically adjusting the shunt of the shunt system, via the at least one shunt drive assembly of the shunt system, relative to at least one cooling coil of the induction furnace and a power coil of the induction furnace during the melting process. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards at least one cooling coil of the induction furnace and the power coil of the induction furnace, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one

signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil of the induction furnace and the power coil of the induction furnace and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least another shunt drive assembly of the shunt system with the shunt of the shunt system; providing at least another load cell with the shunt at a second position and with the at least another shunt drive assembly; and monitoring pressure applied to the shunt by the at least another load cell; wherein the second position is different than the first position. This exemplary embodiment or another exemplary embodiment may further include that the at least one shunt drive assembly and the at least another shunt drive assembly separately and independently automatically adjust the shunt; and wherein the at least one load cell and the at least another load cell of the shunt system separately and independently monitor the shunt. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards at least one cooling coil and a power coil, via at least one of the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via one of the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly and the at least another shunt drive assembly of the shunt system, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil.

[0016] In yet another aspect, an exemplary embodiment of the present disclosure may provide a method of monitoring pressure applied to a head of an induction furnace during a melting process. The method includes steps of engaging at least one head drive assembly of a head system with an apron of the head system and the head of the head system; providing at least one load cell with the head at a first position and with the at least one head drive assembly of the head system; connecting

a controller of the induction furnace with the at least one load cell; connecting the controller with the at least one head drive assembly; and monitoring pressure applied to the head during the melting process by the at least one load cell.

[0017] This exemplary embodiment or another exemplary embodiment may further include a step of automatically adjusting the head of the head system, via the at least one head drive assembly, relative to at least one cooling coil of the induction furnace or a power coil of the induction furnace during the melting process. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and retracting the head towards the apron and towards at least one cooling coil and a power coil, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell of the head system measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from at least one cooling coil and a power coil and away from the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of engaging at least another head drive assembly with the head and the apron; providing at least another load cell with the head at a second position and with the at least another head drive assembly; and monitoring pressure applied to the head by the at least another load cell; wherein the second position is different than the first position. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; and advancing the head towards the apron and towards at least one cooling coil and a power coil, via at least one of the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from the apron and away from the at least one cooling coil and the power coil, via at least one of the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the head; and advancing the head towards the apron and towards the at least one cooling coil and the power coil, via the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. This exemplary embodiment or another exemplary embodiment may further include steps of sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the head; and withdrawing the head away from the apron

and away from the at least one cooling coil and the power coil, via at least one of the at least one head drive assembly and the at least another head drive assembly of the head system, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil.

[0018] In yet another aspect, an exemplary embodiment of the present disclosure may provide a method of monitoring pressure applied to a shunt and a head of an induction furnace during a melting process. The method includes steps of engaging at least one shunt drive assembly of a shunt system with a support column of the shunt system and the shunt of the shunt system; engaging at least one head drive assembly of a head system with an apron of the head system and the head of the head system; engaging at least one cooling coil of the induction furnace and a power coil of the induction furnace with the shunt; engaging the at least one cooling coil of the induction furnace with the head; providing at least one load cell with the shunt and with the at least one shunt drive assembly of the shunt system; providing at least another load cell with the head and with the at least one head drive assembly of the head system; connecting a controller of the induction furnace with the at least one load cell and the at least another load cell; connecting the controller with the at least one shunt drive assembly and the at least one head drive assembly; and monitoring pressure applied to the shunt and the head of the induction furnace during the melting process.

[0019] In yet another aspect, an exemplary embodiment of the present disclosure may provide a coreless induction furnace. The coreless induction furnace includes an outer shell. The coreless induction furnace also includes a set of adjustable shunt systems that operably engages with the outer shell. The coreless induction furnace also includes at least one cooling coil that operably engages with the set of adjustable shunt systems. The coreless induction furnace also includes a power coil that operably engages with the at least one cooling coil and the set of adjustable shunt systems; wherein the set of adjustable shunt systems is configured to automatically adjust at least one shunt of the set of adjustable shunt systems relative to at least one of the at least one cooling coil and the power coil.

[0020] This exemplary embodiment or another exemplary embodiment may further include that each adjustable shunt system of the adjustable shunt systems comprises: a support column operably engaged with the outer shell; and at least one shunt drive assembly operably engaged with the support column and the at least one shunt; wherein the shunt operably engages with the at least one cooling coil and the power coil; and wherein the at least one shunt drive assembly is configured to automatically adjust the shunt along an axis relative to the support column. This exemplary embodiment or another exemplary embodiment may further include that when the shunt is adjusted by the at least one shunt drive assembly, the at least one cooling coil and the power coil are adjusted with the shunt. This exemplary embodiment or another exemplary embodiment may further include that the at least one shunt drive assembly comprises: a first jack having a first end that operably engages with the shunt and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the shunt along the axis that is orthogonal to the support column. This exemplary embodiment or another exemplary embodiment may further include that each adjustable shunt system of the set of adjustable shunt systems further comprises: at least another shunt drive assembly operably engaged with the support column and the shunt; wherein the at least another shunt drive assembly is configured to automatically adjust the shunt along a second axis that is orthogonal to the support column and parallel with the axis of the at least one shunt drive assembly; and wherein each of the at least one shunt drive assembly and the at least another shunt drive assembly adjusts the shunt independently and separately from one another. This exemplary embodiment or another exemplary embodiment may further include that each adjustable shunt system of the set of adjustable shunt systems further comprises: a transition plate operably engaged with the shunt, the at least one cooling coil, and the power coil; wherein the transition plate is positioned between the shunt and the at least one cooling coil and the power coil such that the shunt is spaced apart from the at least

one cooling coil and the power coil. This exemplary embodiment or another exemplary embodiment may further include that each adjustable shunt system of the set of adjustable shunt systems further comprises: a track operably engaged with the support column; and a sled assembly operably engaged with the track and operably engaged with the shunt at one of a first end of the shunt and a second end of the shunt opposite to the first end; wherein the sled assembly is configured to guide the shunt along the track. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the shunt and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; and a controller electrically connected with the at least one load cell and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; wherein the controller is configured to automatically adjust the shunt, via the at least one shunt drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined pressures. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the shunt and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; at least another load cell operably engaged with the shunt and the at least another shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one shunt drive assembly, and the at least another shunt drive assembly; wherein the controller is configured to automatically adjust the shunt, via at least one of the at least one shunt drive assembly and the at least another shunt drive assembly, when at least one of the at least one load cell and the at least another load cell measures pressure that is outside a range of predetermined pressures.

[0021] This exemplary embodiment or another exemplary embodiment may further include an adjustable head system operably engaged with the outer shell; wherein the adjustable head system is configured to automatically adjust at least a head of the adjustable head system relative to the at least one cooling coil and the power coil; and wherein the set of adjustable shunt systems and the adjustable head system are automatically adjustable independently and separately from one another. This exemplary embodiment or another exemplary embodiment may further include that the adjustable head system comprises: a head; an apron operably engaged with the outer shell; and at least one head drive assembly operably engaged with the apron; wherein the at least one head drive assembly is configured to automatically adjust the head along an axis relative to the apron. This exemplary embodiment or another exemplary embodiment may further include that when the head is adjusted by the at least one head drive assembly, the at least one cooling coil and the power coil are adjusted with the head. This exemplary embodiment or another exemplary embodiment may further include that the at least one head drive assembly comprises: a first jack having a first end that operably engages with the head and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the head along the axis relative to the apron. This exemplary embodiment or another exemplary embodiment may further include that the adjustable head system further comprises: at least another head drive assembly operably engaged with the apron and the head; wherein the at least another head drive assembly is configured to automatically adjust the head along a second axis relative to the apron and parallel with the axis of the at least one head drive assembly; and wherein each of the at least one head drive assembly and the at least another head drive assembly adjusts the head independently and separately from one another. This exemplary embodiment or another exemplary embodiment may further include that the head of the adjustable head system further comprises: a top plate operably engaged with the at least one head drive assembly; and a refractory member operably engaged with the top plate and configured to engage with the at least one cooling coil to maintain a desired pressure on the at least one cooling coil and the power coil; wherein the top plate and the refractory member are adjustable relative to the apron and the at least one cooling coil. This exemplary embodiment or another exemplary embodiment may further include at least

one load cell operably engaged with the head and the at least one head drive assembly; and a controller electrically connected with the at least one load cell and the at least one head drive assembly; wherein the controller is configured to automatically adjust the head, via the at least one head drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined pressures. This exemplary embodiment or another exemplary embodiment may further include at least one load cell operably engaged with the head and the at least one head drive assembly; at least another load cell operably engaged with the head and the at least another head drive assembly; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one head drive assembly, and the at least another head drive assembly; wherein the controller is configured to automatically adjust the shunt, via at least one of the at least one head drive assembly and the at least another head drive assembly, when at least one of the at least one load cell and the at least another load cell measure a pressure that is outside a range of predetermined pressures.

[0022] In yet another aspect, an exemplary embodiment of the present disclosure may provide a method of automatically adjusting one of a shunt of a shunt system and a head of a head system of a coreless induction furnace. The method includes steps of engaging the shunt system with an outer shell of the coreless induction furnace; engaging the head system with the outer shell of the coreless induction furnace; engaging at least one cooling coil of the coreless induction furnace with the shunt system and the head system; engaging a power coil of the coreless induction furnace with the shunt system and the at least one cooling coil; and automatically adjusting one of the shunt of the shunt system and a head of the head system of the coreless induction furnace.

[0023] This exemplary embodiment or another exemplary embodiment may further include that the steps of engaging the shunt system with the outer shell, the at least one cooling coil, and the power coil further comprises: engaging a support column of the shunt system with the outer shell; engaging at least one shunt drive assembly of the shunt system with the support column of the shunt system; engaging the at least one shunt drive assembly with the shunt of the shunt system; engaging the at least one cooling coil of the coreless induction furnace and the power coil of the coreless induction furnace with the shunt; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least one shunt drive assembly, relative to the at least one cooling coil and the power coil. This exemplary embodiment or another exemplary embodiment may further include that steps of engaging the head system with the outer shell and the at least one cooling coil further comprises: engaging an apron of the shunt system with the outer shell; engaging at least one head drive assembly of the head system with the apron of the head system; engaging the at least one head drive assembly with the head of the head system; engaging at least one cooling coil of the coreless induction furnace with the head; and automatically adjusting the head of the head system of the coreless induction furnace, via the at least one head drive assembly, relative to the at least one cooling coil.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0024] Sample embodiments of the present disclosure are set forth in the following description, are shown in the drawings and are particularly and distinctly pointed out and set forth in the appended claims.

[0025] FIG. 1 (FIG. 1) is a partial sectional view of an induction furnace in accordance with one aspect of the present disclosure, wherein the induction furnace includes adjustable shunt systems and an adjustable head system.

[0026] FIG. 2 (FIG. 2) is a top plan view of the induction furnace taken in the direction of line 2-2 shown in FIG. 1.

[0027] FIG. 3 (FIG. 3) is a sectional view of the induction furnace taken in the direction of line 3-3 shown in FIG. 1.

[0028] FIG. 4 (FIG. 4) is a sectional view of the induction furnace taken in the direction of line 4-4 shown in FIG. 1.

[0029] FIG. 4A (FIG. 4) is an enlarged view of the highlighted region shown in FIG. 4.

[0030] FIG. 5 (FIG. 5) is a partial isometric perspective view of at least one adjustable shunt system of the adjustable shunt systems of the induction furnace shown in FIG. 1.

[0031] FIG. 6 (FIG. 6) is a partial cross-sectional view taken in direction of line 6-6 shown in FIG. 5.

[0032] FIG. 7 (FIG. 7) is a partial isometric perspective view of at least one adjustable shunt system of the adjustable shunt systems of the induction furnace and the adjustable head system of the induction furnace shown in FIG. 1.

[0033] FIG. 8 (FIG. 8) is an enlarged view of the highlighted region shown in FIG. 1.

[0034] FIG. 9 (FIG. 9) is an enlarged view of the highlighted region shown in FIG. 1.

[0035] FIG. 10 (FIG. 10) is an operational view of the induction furnace during a melting process, wherein the head of the adjustable head system and a shunt of at least one adjustable system may be adjusted in response to pressure exerted on the head or the shunt.

[0036] FIG. 11A (FIG. 11A) is an exemplary flowchart of a set of instructions for a controller of induction furnace shown in FIG. 1.

[0037] FIG. 11B (FIG. 11B) is an exemplary flowchart of another set of instructions for a controller of induction furnace shown in FIG. 1.

[0038] FIG. 12A (FIG. 12A) is an exemplary flowchart of another set of instructions for a controller of induction furnace shown in FIG. 1.

[0039] FIG. 12B (FIG. 12B) is an exemplary flowchart of another set of instructions for a controller of induction furnace shown in FIG. 1.

[0040] FIG. 13 (FIG. 13) is an exemplary method flowchart.

[0041] FIG. 14 (FIG. 14) is another exemplary method flowchart.

[0042] FIG. 15 (FIG. 15) is another exemplary method flowchart.

[0043] FIG. 16 (FIG. 16) is another exemplary method flowchart.

[0044] FIG. 17 (FIG. 17) is another exemplary method flowchart.

[0045] FIG. 18 (FIG. 18) is another exemplary method flowchart.

[0046] Similar numbers refer to similar parts throughout the drawings.

DETAILED DESCRIPTION

[0047] FIGS. 1-10 illustrate a coreless induction furnace or induction furnace, which is generally referred to as 1. It should be appreciated that induction furnace 1 may be configured to apply electrical energy to induce induction heating of various types of metals provided inside induction furnace 1. Induction furnace 1 described and illustrated herein, however, is configured to automatically adjust or move the positions of at least one shunt provided in induction furnace 1 and/or a head of induction furnace when assembling induction furnace 1, disassembling induction furnace 1, or when melting a metal material in induction furnace 1; such assemblies and/or components or adjusting at least one shunt and/or a head of induction furnace 1 are described in greater detail below.

[0048] It should be understood that while induction furnace 1 is a coreless induction furnace, such systems, assemblies, and components of induction furnace 1 may be included in other various types of induction furnaces, including channel induction furnaces and other various types of induction furnaces of the like.

[0049] Induction furnace 1 may include a top end 1A, a bottom end 1B vertically opposite to top end 1A, and a vertical axis Z defined therebetween. Induction furnace 1 may also include a first or front side 1C, a second or rear side 1D longitudinally opposite to the front end 1C, and a longitudinal axis Y defined therebetween (see FIG. 2). Induction furnace 1 may also include a third

or left side **1E**, a fourth or right side **1F** transversely opposite to left side **1E**, and a transverse axis **X** defined therebetween.

[0050] Induction furnace **1** may include an outer shell or main support **10**. Outer shell **10** is configured to be the main support mechanism and/or foundation of the assemblies and/or components housed inside of induction furnace **1**. Outer shell **10** is also configured to be the final flux shield between a power coil of induction furnace **1** and the exterior environment surrounding the induction furnace **1** for preventing any stray flux from leaving the induction furnace during melting properties.

[0051] As best seen in FIG. **1**, outer shell **10** may include a top end **10A**, a bottom end **10B** vertically opposite to top end **10A**, and a vertical axis defined therebetween parallel with vertical axis **Z**. Outer shell **10** may also include a circumferential wall **10C** that extends between top end **10A** and bottom end **10B** and defines a cylindrical-shape. Outer shell **10** may also define a chamber **10D** by the circumferential wall **10C** and extends between the top end **10A** and the bottom end **10B**. As best seen in FIG. **1**, outer shell **10** may also include a base wall or support **10E** that extends from the circumferential wall **10C** and into the chamber **10D**; such use and purpose of base wall **10E** is described in more detail below.

[0052] It should be appreciated the outer shell **10** may be made from any suitable material that is configured to be the main support mechanism and/or foundation of the assemblies and/or components housed inside of induction furnace **1** as well as being the final flux shield between a power coil of induction furnace **1** and the exterior environment surrounding the induction furnace **1**. In one exemplary embodiment, outer shell described and illustrated herein may be formed from a metal material, particularly steel or a similar metal material.

[0053] Induction furnace **1** may also include at least one adjustable shunt system **20** that operably engages with the outer shell **10** inside of chamber **10D**. More particularly, the at least one adjustable shunt system **20** operably engages with base wall **10E** of outer shell **10** inside of chamber **10D**. In the illustrated embodiment, induction furnace **1** may include a plurality of adjustable shunt systems **20** that operably engages with the outer shell **10** inside of chamber **10D** (see FIG. **4**). In this illustrated embodiment, induction furnace **1** may include twelve adjustable shunt systems **20** that operably engages with the outer shell **10** inside of chamber **10D**. In other exemplary embodiments, any suitable number of adjustable shunt systems may be provided in an induction furnace based on various considerations, include the size, shape, and configuration of the induction furnace. As described in more detail below, the shunt in each adjustable shunt system **20** of induction furnace **1** is configured to be automatically adjustable in response to pressure exerted on the shunt during assembly operations, disassembling operations, and melting processes. Such assemblies and components of each adjustable shunt system **20** are described in more detail below.

[0054] Each adjustable shunt system **20** may include a support column **22** that may operably engage with outer shell **10**. More particularly, support column **22** may operably engage with the base wall **10E** of outer shell **10** to maintain the support column **22** inside of chamber **10D** of outer shell **10** (as shown in FIG. **1**). As best seen in FIG. **1**, support column **22** may include a top end **22A** proximate to the top end **10A** of outer shell **10**, a bottom end **22B** vertically opposite to the top end **22A** and operably engaged with the base wall **10E**, and a vertical axis defined therebetween parallel with vertical axis **Z**. Support column **22** may also include an outer surface **22C** that extends between the top end **22A** and the bottom end **22B** (best seen in FIG. **5**). Support column **22** may also include an inner surface **22D** that extends between the top end **22A** and the bottom end **22B**; the inner surface **22D** is positioned interior to the outer surface **22C** and faces in an opposite direction away from the outer surface **22C**. Support column **22** may also define a chamber **22E** that is defined between the top end **22A** and the bottom end **22B** in which the inner surface **22D** is positioned inside of chamber **22E**. Such use and purpose of the chamber **22E** defined by support column **22** is described in more detail below.

[0055] Support column **22** of each adjustable shunt system **20** may also define at least one pair of

apertures where each aperture of the at least one pair of aperture are coaxial with one another. As best seen in FIGS. 1 and 5, support column 22 may define a first or upper pair of apertures 22F where each aperture of the upper pair of apertures 22F is coaxial with one another. Each aperture of the upper pair of apertures 22F also extends entirely through the support column 22 where the outer surface 22C and the inner surface 22D are in fluid communication with one another via the upper pair of apertures 22F. Similarly, support column 22 may define a second or lower pair of apertures 22G where each aperture of the lower pair of apertures 22G is coaxial with one another. Each aperture of the lower pair of apertures 22G also extends entirely through the support column 22 where the outer surface 22C and the inner surface 22D are in fluid communication with one another via the lower pair of apertures 22G. Such uses and purposes of the upper pair of apertures 22F and the lower pair of apertures 22G are described in more detail below.

[0056] Support column 22 may define a first side aperture 22H where first side aperture 22H is defined between the upper pair of apertures 22F. First side aperture 22H also extends entirely through the support column 22 where the outer surface 22C and the inner surface 22D are in fluid communication with one another via first side aperture 22H. While not illustrated herein, support column 22 may define a second side aperture (not illustrated) where second side aperture is defined between the lower pair of apertures 22G. Second side aperture also extends entirely through the support column 22 where the outer surface 22C and the inner surface 22D are in fluid communication with one another via second side aperture. Such uses and purposes of the first and second side apertures 22H are described in more detail below.

[0057] Each adjustable shunt system 20 may also include at least one shunt drive assembly 24 that operably engages with the support column 22. As described in more detail below, the at least one shunt drive assembly 24 in each adjustable shunt system 20 is configured to move a respective shunt in response to pressure applied against the shin during assembling operations of induction furnace 1, disassembling operations of induction furnace 1, or melting processes performed in induction furnace 1. The at least one shunt drive assembly 24 may also be operably engaged with the support column 22 inside of the chamber 22E. In the illustrated embodiment, each adjustable shunt system 20 of induction furnace 1 may include a first shunt drive assembly 24A and a second shunt drive assembly 24B that may be operably engaged with the support column 22 inside of the chamber 22E and are substantially similar to one another. It should be appreciated that while first shunt drive assembly 24A and second shunt drive assembly 24B are substantially similar to one another, first shunt drive assembly 24A will be described in more detail below for brevity. Inasmuch as the first shunt drive assembly 24A is described, it should be appreciated that the description of the second shunt drive assembly 24B is substantially similar to the first shunt drive assembly 24A.

[0058] First shunt drive assembly 24A may include a motor 26 and a jack 28 that operably engages with the motor 26. As best seen in FIG. 5, motor 26 may define a passageway 26A that extends entirely through motor 26 to enable jack 28 to pass through motor 26 when motor 26 linearly moves jack 28 during operation. Referring to FIG. 1, jack 28 may include a first end 28A and a second end 28B longitudinally opposite to the first end 28A and operably engages with a shunt of adjustable shunt system 20, which is described in more detail below. Motor 26 may also be operably engaged with support column 22 inside of chamber 22E. More particularly, motor 26 may be operably engaged with the inner surface 22D of support column 22 inside of chamber 22E. In other exemplary embodiments, motor 26 may be operably engaged with support column 22 in various structural configurations such that the motor 26 is maintained with support column 22.

[0059] During operation, motor 26 is operable to linearly move the jack 28 inwardly and outwardly of support column 22 to automatically adjust a shunt of adjustable shunt system 20. Stated differently, motor 26 is operable to linearly move the jack 28 inwardly and outwardly of support column 22 along an axis that is orthogonal to the longitudinal axis of support column 22. Such linear movement of jack 28 by motor 26 is denoted by a double arrow labeled "LM1" in FIG. 4A.

It should be appreciated that each aperture of the upper pair of apertures 22F is also sized and configured to enable the jack 28 to pass through the support column 22 when the motor 26 linearly moves the jack 28 inwardly and outwardly of support column 22.

[0060] In the illustrated embodiment, first shunt drive assembly 24A is a motorized jackscrew mechanism for automatically adjusting a shunt of adjustable shunt system 20. In this illustrated embodiment, motor 26 is configured to rotate jack 28 about the longitudinal axis of jack 28 in order to linearly move the jack 28 inwardly and outwardly relative to support column 22; such rotational force applied to jack 28 via motor 26 is denoted by an arrow labeled “RM1” in FIG. 10. In other exemplary embodiments, any suitable mechanical mechanism and/or assembly may be used for automatically adjust a shunt of adjustable shunt system 20, including, electrical mechanisms, hydraulic mechanisms, or pneumatic mechanisms for automatically adjusting or moving a shunt in response to extreme pressure applied to the shunt.

[0061] As discussed previously, first shunt drive assembly 24A and second shunt drive assembly 24B operably engaged with the support column 22 inside of the chamber 22E and are substantially similar to one another. As such, motor 26 and jack 28 of second shunt drive assembly 24B are substantially similar to motor 26 and jack 28 of first shunt drive assembly 24A previously described above. It should also be appreciated that each aperture of the lower pair of apertures 22G is also sized and configured to enable the jack 28 of second shunt drive assembly 24B to pass through the support column 22 when the motor 26 of second shunt drive assembly 24B linearly moves the jack 28 inwardly and outwardly of support column 22.

[0062] Each adjustable shunt system 20 may also include a shunt 30 that operably engages with at least one shunt drive assembly 24. In the illustrated embodiment, shunt 30 of each adjustable shunt system 20 may operably engage with first shunt drive assembly 24A at a first position and with second shunt drive assembly 24B at a second position that is different than the first position. As described in more detailed below, shunt 30 is moveable and/or adjustable relative to the support column 22 along an axis that is orthogonal to the longitudinal axis of support column 22 by one or both of the first shunt drive assembly 24A and the second shunt drive assembly 24B in response to relieving pressure from the shunt 30 or applying pressure to the shunt 30; such operations of relieving pressure from shunt 30 and applying pressure to shunt 30 are described in more detail below.

[0063] As best seen in FIGS. 4A and 5, shunt 30 of each adjustable shunt system 20 may include a top end 30A proximate to top end 22A of support column 22, a bottom end 30B vertically opposite to top end 30A and proximate to bottom end 22B of support column 22, and a longitudinal axis defined therebetween parallel with vertical axis Z of induction furnace 1. Shunt 30 may also be formed by a series of laminate sheets 30C that is compressed together via a support bracket 30D. The series of laminate sheets 30C is also secured with the support bracket 30D via at least one securement mechanism 30E (e.g. a bolt threadably engaged with a nut). As best seen in FIGS. 4A and 5, shunt 30 may include a plurality of securement mechanisms 30E that secure the series of laminate sheets 30C with the support bracket 30D. It should be appreciated that while shunt 30 is described and illustrated herein with a particular structural configuration, any suitable, commercially available shunt or yoke may be used in each adjustable shunt system 20 of induction furnace 1 described and illustrated herein.

[0064] Upon assembly of each adjustable shunt system 20, at least one shunt drive assembly 24 may operably engage with the shunt 30 for automatically moving and/or adjusting the shunt 30 in response to relieving pressure from the shunt 30 or applying pressure to the shunt 30. More particularly, shunt 30 is moveable and/or adjustable relative to the support column 22 by one or both of the first shunt drive assembly 24A and the second shunt drive assembly 24B in response to relieving pressure from the shunt 30 or applying pressure to the shunt 30. Specifically, jacks 28 of the first shunt drive assembly 24A and the second shunt drive assembly 24B may be engaged with one or both of the series of laminate sheets 30C and the support bracket 30D for moving and/or

adjusting the shunt **30** in response to relieving pressure from the shunt **30** or applying pressure to the shunt **30**.

[0065] Still referring to FIGS. **4A** and **5**, shunt **30** may define a first side surface **30F** extending along the entire length of shunt **30** between the top end **30A** and the bottom end **30B**. In the illustrated embodiment, first side surface **30F** is substantially planar along the entire length of shunt **30** between the top end **30A** and the bottom end **30B**. Shunt **30** may also define a second side surface **30G** extending along the entire length of shunt **30** between the top end **30A** and the bottom end **30B**. As best seen in FIGS. **4A** and **5**, second side surface **30G** is also transversely opposite to the first side surface **30F**, and the first side surface **30F** and the second side surface **30G** face in opposing directions. In the illustrated embodiment, second side surface **30G** is also substantially planar along the entire length of shunt **30** between the top end **30A** and the bottom end **30B**. Such use of the first side surface **30F** and the second side surface **30G** being substantially planar surface is described in more detail below.

[0066] Referring to FIG. **4A**, shunt **40** may also define sets of openings **30H** that extend entirely through each laminate sheet of the series of laminate sheets **30C** and support bracket **30D**. With such sets of openings **30H**, the bolts of the plurality of securement mechanisms **30E** may pass through the series of laminate sheets **30C** and support bracket **30D** to hold the series of laminate sheets **30C** and support bracket **30D** with one another.

[0067] Induction furnace **1** may also include a transition plate **40** with each adjustable shunt system **20**. As best seen in FIGS. **4A** and **5**, transition plate **40** of each adjustable shunt system **20** operably engages with the shunt **30** of each adjustable shunt system **20**. Transition plate **40** may also be positioned between the shunt **30** of each adjustable shunt system **20** and one of both of at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1**, which are described in greater detail below. Such features and characteristics of transition plate **40** of each adjustable shunt system **20** are described in greater detail below.

[0068] As best seen in FIGS. **4A** and **5**, transition plate **40** may include a top end **40A** proximate to top end **30A** of shunt **30**, a bottom end **40B** vertically opposite to top end **40A** and proximate to bottom end **30B** of shunt **30**, and a longitudinal axis defined therebetween parallel with vertical axis **Z** of induction furnace **1**. Transition plate **40** may also define a first side surface **40C** extending along the entire length of transition plate **40** between the top end **40A** and the bottom end **40B**. In the illustrated embodiment, first side surface **40C** is substantially planar along the entire length of transition plate **40** between the top end **40A** and the bottom end **40B**. Transition plate **40** may also define a second side surface **40D** extending along the entire length of transition plate **40** between the top end **40A** and the bottom end **40B**. As best seen in FIGS. **4A** and **5**, second side surface **40D** is also transversely opposite to the first side surface **40C**, and the first side surface **40C** and the second side surface **40D** face in opposing directions. In the illustrated embodiment, second side surface **40D** is curvilinear-shaped or is concavely-shaped along the entire length of transition plate **40** between the top end **40A** and the bottom end **40B**.

[0069] Transition plate **40** may also include a third side surface **40E** and a fourth side surface **40F** opposite to the third side surface **40E**. As best seen in FIGS. **4A-5**, third side surface **40E** may extend between top end **40A** and bottom end **40B** and positioned intermediate the first side surface **40C** and the second side surface **40D**. Still referring to FIGS. **4A-5**, fourth side surface **40F** may also extend between top end **40A** and bottom end **40B** and positioned intermediate the first side surface **40C** and the second side surface **40D** opposite to the third side surface **40E**. As best seen in FIG. **4A**, first side surface **40C** may define a first width **40G** that is measured between the third side surface **40E** and the fourth side surface **40F**. Similarly, second side surface **40D** may also define a second width **40H** that is measured between the third side surface **40E** and the fourth side surface **40F** where the second width **40H** is greater than the first width **40G**.

[0070] In other exemplary embodiments, first side surface **40C** and second side surface **40D** may define any suitable widths based on the structure configuration of transition plate **40**. In one

exemplary embodiment, first width **40G** of first side surface **40C** may be greater than second width **40H** of second side surface **40D**. In another exemplary embodiment, first width **40G** of first side surface **40C** may be equal with second width **40H** of second side surface **40D**.

[0071] Upon assembly of induction furnace **1**, transition plate **40** is configured to engage with the respective shunt in each adjustable shunt system **20** while engaging with one of both of at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1**. As best seen in FIGS. **4A** and **5**, the first side surface **40C** of the transition plate **40** directly contacts and engages with the second side surface **30G** of shunt **30**, and the second side surface **40D** of the transition plate **40** engages with one or both of at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1**. As such, the transition plate **40** of each adjustable shunt system **20** is positioned between the shunt **30** of each adjustable shunt system **20** and one of both of at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1**.

[0072] The structural configuration and positioning of the transition plate **40** in each adjustable shunt system **20** is considered advantageous at least because the transition plate **40** promotes uniform and even pressure to be applied to at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1** via the shunt **30**. In the illustrated embodiment, the engagement between the planar surface **30G** of shunt **30** and planar surface **40C** of transition plate **40** enables uniform and even pressure to be transferred between shunt **30** and transition plate **40** when pressured is applied by shunt **30**, via one or both of the first and second shunt drive assemblies **24A**, **24B**, or is applied by transition plate **40** via a power coil of induction furnace **1**.

[0073] The structural configuration and positioning of the transition plate **40** in each adjustable shunt system **20** is also considered advantageous at least because the transition plate **40** may enable the shunt **30** to provide greater force and/or pressure on at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1** by providing leverage on the transition plate **40** via the shunt **30**. Moreover, such structural configuration and positioning of the transition plate **40** may prevent warpage and/or deterioration of shunt **30** when a power coil of induction furnace **1** exerts force during melting operations. Such force exerted by a power coil of induction furnace **1** will be applied to the curvilinear shape of transition plate **40** (i.e., second side surface **40D**), which matches and/or is complementary with the shape a power coil of induction furnace **1**. As such, shunt **30** of each adjustable shunt system **20** is isolated from a power coil of induction furnace **1** such that transition plate **40** of each adjustable shunt system **20** acts a barrier or absorber of pressure exerted by the power coil of induction furnace **1**.

[0074] Since transition plate **40** of each adjustable shunt system **20** engages with a power coil of induction furnace **1**, transition plate **40** of each adjustable shunt system **20** may be made from any suitable material to withstand high level of electrical current and force from a power coil along with extreme heat emitted from the molten charge melting inside of induction furnace **1** during melting processes. More particularly, transition plate **40** of each adjustable shunt system **20** may be formed from a composite material for withstanding high level of electrical current and force from a power coil along with extreme heat emitted from the molten charge melting inside of induction furnace **1** during melting processes. Specifically, transition plate **40** of each adjustable shunt system **20** may be formed from a high-pressure fiberglass laminate, such as G-10 or garolite, for withstanding high level of electrical current and force from a power coil along with extreme heat emitted from the molten charge melting inside of induction furnace **1** during melting processes.

[0075] Each adjustable shunt system **20** may also include a track **50** and a sled assembly that slidably engages with the track **50**. As best seen in FIGS. **5** and **6**, track **50** of each adjustable shunt system **20** may operably engage with the outer shell **10** inside of chamber **10D**, particularly with the base wall **10E** of outer shell **10**. Optionally, track **50** of each adjustable shunt system **20** may also operably engage with the support column **22** of each adjustable shunt system **20** while also being engaged with the outer shell **10**. Still referring to FIGS. **5** and **6**, sled assembly **60** of each adjustable shunt system **20** may also operably engage with the shunt **30** where the shunt **30** may be

guided along the track **50** by assistance of sled assembly **60**. Such features of track **50** along with components of sled assembly **60** are described in further detail below.

[0076] As best seen in FIGS. **5** and **6**, track **50** of each adjustable shunt system **20** may include a first end **50A** proximate to the support column **22**, a second end **50B** longitudinally opposite to the first end **50A** and remote from support column **22**, and a longitudinal axis defined therebetween. Track **50** may also include an upper horizontal wall **50C** that extends longitudinally between the first end **50A** and the second end **50B**. Upper horizontal wall **50C** may define a top surface **50C1** and a bottom surface **50C2** vertically opposite to the top surface **50C1** and facing away from the top surface **50C1**. Track **50** may also include a vertical wall **50D** that extends orthogonally from the upper horizontal wall **50C**, particularly at the bottom surface **50C2**, and along an axis that is orthogonal to the longitudinal axis of track **50**. Track **50** may also include a lower horizontal wall **50E** that operably engages with the vertical wall **50D**. Similar to upper horizontal wall **50C**, lower horizontal wall **50E** also extends longitudinally between the first end **50A** and the second end **50B** such that the upper horizontal wall **50C** and the lower horizontal wall **50E** are parallel with one another and is orthogonal with the vertical wall **50D**. With such configuration, track **50** may define a l-shaped cross section where track **50** may be an I-beam member.

[0077] Still referring to FIGS. **5** and **6**, sled assembly **60** may include a frame **62**. Frame **62** includes a first upright wall **62A**, a second upright wall **62B** positioned opposite to the first upright wall **62A**, and a connecting wall **62C** operably engaged with the first upright wall **62A** and the second upright wall **62B** where the connecting wall **62C** is positioned between the first upright wall **62A** and the second upright wall **62B**. First upright wall **62A** may also define a first set of openings **62A1** where each opening of the first set of openings **62A1** extends entirely through the first upright wall **62A** along an axis that is parallel with the connecting wall **62C**. Similarly, second upright wall **62B** may also define a second set of openings **62B1** where each opening of the second set of openings **62B1** extends entirely through the second upright wall **62B** along an axis that is parallel with the connecting wall **62C**. It should be appreciated that the first set of openings **62A1** and the second set of openings **62B1** are coaxial with one another. With such configuration, sled assembly **60** may define a U-shaped cross section that may partially house the track **50** inside of the frame **62** (best seen in FIG. **6**). More particularly, upper horizontal wall **50C** and a portion of vertical wall **50D** may be housed inside of frame **62**.

[0078] Still referring to FIGS. **5** and **6**, sled assembly **60** may also include at least one roller assembly that operably engages with the frame **62** to slidably move and guide the frame **62** and the shunt **30** along the track **50**. In the illustrated embodiment, sled assembly **60** includes a set of roller assemblies **64** that operably engages with the frame **62** to slidably move and guide the frame **62** and the shunt **30** along the track **50**; such sliding engagement and movement of the sled assembly **60** along track **50** is denoted by a double arrow labeled “SE” in FIG. **5**. Each roller assembly of the set of roller assemblies **64** may include at least one roller or wheel **65A**, at least one spacer sleeve **65B** operably engaged between one of the first upright wall **62A** and the second upright wall **62C** and the at least one wheel **65A**, and a bolt **65C** threadably engaged with a nut **65D** for securing the at least one wheel **65A** and the at least one spacer sleeve **65B** with the frame **62**. Bolt **65C** is also operably engaged with the frame **62** and configured to enable the at least one wheel **65A** to rotate about a longitudinal axis of the bolt **65C**. As best seen in FIG. **6**, bolt **65C** of each roller assembly of the set of roller assemblies **64** may pass through one of the openings in the first set of openings **62A1** defined in the first upright wall **62A** or in the second set of openings **62B1** defined in the second upright wall **62B**.

[0079] It should be appreciated that any suitable number of roller assemblies from the set of roller assemblies **64** may slidably engage with the track **50**. In one example, at least one roller assembly of the set of roller assemblies **64** may slidably engage with the top surface **50C1** of upper horizontal wall **50C** of track **50**, and at least another roller assembly of the set of roller assemblies **64** may slidably engage with the bottom surface **50C2** of upper horizontal wall **50C** of track **50**. In

another example, a first group of roller assemblies **64A** may slidably engage with the top surface **50C1** of upper horizontal wall **50C** of track **50**, a second group of roller assemblies **64B** may slidably engage with the bottom surface **50C2** of upper horizontal wall **50C** of track **50** and operably engage with the first upright wall **62A**, and a third group of roller assemblies **64C** may slidably engage with the bottom surface **50C2** of upper horizontal wall **50C** of track **50** and operably engage with the second upright wall **62B** (see FIG. 6).

[0080] It should also be appreciated that track **50** and sled assembly **60** may be positioned at any suitable location relative to shunt **30** while still being able to slidably move and guide the shunt **30** along track **50**. In one instance, track **50** and sled assembly **60** may be positioned vertically above the shunt **30** such that the track **50** is vertically above the top end **30A** of shunt **30** and the sled assembly **60** is operably engaged at the top end **30A** of shunt **30**. In another instance, track **50** and sled assembly **60** may be positioned vertically below the shunt **30** such that the track **50** is vertically below the bottom end **30B** of shunt **30** and the sled assembly **60** is operably engaged at the bottom end **30B** of shunt **30** (see FIGS. 5 and 6). In yet another instance, track **50** and sled assembly **60** may be positioned between the top end **30A** and the bottom end **30B** of the shunt **30** such that the track **50** is positioned between the top end **30A** and the bottom end **30B** of the shunt **30** and the sled assembly **60** is operably engaged with the shunt **30** at a position between the top end **30A** and the bottom end **30B** of the shunt **30**.

[0081] While not illustrated herein, sled assembly **60** may be configured with a housing that holds and seals devices and/or components for powering electrical components provided in induction furnace **1**, including shunt adjust assemblies **24A**, **24B**. As such, the housing would translate and/or move with shunt **30** during assembly operations of induction furnace **1**, disassembly operations of induction furnace **1**, and melting processes performed by induction furnace **1**.

[0082] Induction furnace **1** may also include an adjustable head system **70** that operably engages with the outer shell **10**. During operation, adjustable head system **70** may be configured to automatically adjust a head of adjustable head system **70** to maintain at least one cooling coil of induction furnace **1** and a power coil of induction furnace **1** at a desired pressure when assembling induction furnace **1**, disassembling induction furnace **1**, and melting a molten charge inside of induction furnace **1**. It should be appreciated that adjustable head system **70** may also be configured to automatically adjust a head of the adjustable head system **70** separately and independently from each adjustable shunt system **20** automatically adjust a shunt **30** in response to pressure exerted on head of adjustable head system **70** or shunt **30** of any of the adjustable shunt systems **20**. Such component and assemblies of adjustable head system **70** are described in more detail below.

[0083] Adjustable head system **70** may include an apron **72** that operably engages with the adjustable shunt systems **20**. As best seen in FIG. 3, apron **72** may define be annular-shaped such that the apron **72** matches with the overall shape of outer shell **10**. As best seen in FIGS. 1 and 7, apron **72** may operably engage with the support columns **22** of adjustable shunt systems **20**. Apron **72** may include a top end **72A** positioned remote from support columns **22**, a bottom end **72B** positioned vertically opposite to the top end **72A**, and a surrounding wall **72C** defining a U-shaped cross section that extends between the top end **72A** and the bottom end **72B** and operably engages with the support columns **22**. Apron **72** may also include a cross member **72D** that operably engages with the surrounding wall **72C** between the top end **72A** and the bottom end **72B**. Apron **72** may also define a chamber **72E** between the surrounding wall **72C** and the cross member **72D**. Apron **72** may also define a recess **72F** that extends downwardly from the top end **72A** to the cross member **72D**. Apron **72** may also define a set of openings **72G** in the cross member **72D** that provide fluid communication between the chamber **72E** and the recess **72F**; such use of this set of openings **72G** is described in more detail below.

[0084] Apron **72** may also define a central opening **72H**, via the surrounding wall **72C**, that extends entirely through the apron **72** and is interior of the surrounding wall **72C**. It should be appreciated that chamber **72E** and central opening **72H** are separate and spaced apart from one another via the

surrounding wall **72C**. It should be appreciated that recess **72F** and central opening **72H** are separate and spaced apart from one another via the surrounding wall **72C**.

[0085] Adjustable head system **70** may also include at least one head drive assembly **74** that operably engages with the apron **72**. As best seen in FIG. 7, the at least one head drive assembly **74** operably engages with the apron **72** inside of the chamber **72E**. More particularly, the at least one head drive assembly **74** operably engages with the cross member **72D** of apron **72** inside of the chamber **72E**. In the illustrated embodiment, adjustable head system **70** includes a set of head drive assemblies **74** where each head drive assembly **74** operably engages with the apron **72** inside of the chamber **72E**, more particularly with the cross member **72D** of apron **72** inside of the chamber **72E**. Such components of each head drive assembly in the set of head drive assemblies **74** are described in greater detail below.

[0086] As best seen in FIG. 7, each head drive assembly **74** may include a motor **76** and a jack **78** that operably engages with the motor **76**. As best seen in FIG. 7, motor **76** may define a chamber **76A** that extends into motor **26** to enable jack **78** to pass in and out of motor **76** when motor **76** linearly moves jack **78** during operation. Referring to FIG. 7, jack **78** may include a first end **78A** and a second end **78B** longitudinally opposite to the first end **78A** and operably engages with a head of adjustable head system **70**, which is described in more detail below. Motor **76** may also be operably engaged with apron **72** inside of chamber **72E**. More particularly, motor **76** may be operably engaged with the cross member **72D** of apron **72** inside of chamber **72E**. In other exemplary embodiments, motor **76** may be operably engaged with apron **72** in various structural configurations such that the motor **76** is maintained with apron **72**.

[0087] During operation, motor **76** is operable to linearly move the jack **78** inwardly and outwardly of apron **72** to automatically adjust a head of adjustable head system **70**. Stated differently, motor **76** is operable to linearly move the jack **78** inwardly and outwardly of apron **72** along an axis that is orthogonal to the longitudinal axis of apron **72**. Such linear movement of jack **78** by motor **76** is denoted by a double arrow labeled “LM2” in FIG. 7. It should be appreciated that each opening of the set of openings **72G** is also sized and configured to enable the jack **78** to pass through the apron **72** when the motor **76** linearly moves the jack **78** inwardly and outwardly of apron **72**.

[0088] In this illustrated embodiment, motor **76** is configured to rotate jack **78** about the longitudinal axis of jack **78** in order to linearly move the jack **78** inwardly and outwardly relative to apron **72**; such rotational force applied to jack **78** via motor **76** is denoted by an arrow labeled “RM2” in FIG. 10. In other exemplary embodiments, any suitable mechanical mechanism and/or assembly may be used for automatically adjust a head of adjustable head system **70**, including, electrical mechanisms, hydraulic mechanisms, or pneumatic mechanisms for automatically adjusting or moving a head in response to extreme pressure applied to the head.

[0089] Adjustable head system **70** may also include a head **80** that operably engages with the set of head drive assemblies **74**. As described in more detailed below, head **80** is also moveable and/or adjustable relative to the apron **72** along an axis that is orthogonal to the longitudinal axis of apron **72** by at least one of head drive assemblies **74** in response to relieving pressure from head **80** or applying pressure to head **80**.

[0090] As best seen in FIG. 7, head **80** may include a top plate **82** that operably engages with the set of head drive assemblies **74**. More particularly, top plate **82** operably engages with the jacks **78** of the set of head drive assemblies **74**. Top plate **82** may include a top surface **82A** facing away from chamber **10D**, a bottom surface **82B** vertically opposite to the top surface **82A** and facing into chamber **10D**, and a retaining arm **82C** extending from the bottom surface **82B** and into the chamber **10D**. Head **80** may also include a refractory member **84** that operably engages with the top plate **82** via the retaining arm **82C**. It should be appreciated that refractory member **84** may operably engage with at least one cooling coil of induction furnace **1** to maintain a desired pressure on at least one cooling coil of induction furnace **1** as well as a power coil of induction furnace **1**. Such engagement of the head **80** with at least one cooling coil of induction furnace **1** as well as a

power coil of induction furnace **1** is described in more detail below.

[0091] Induction furnace **1** may also include a power coil **90**. As best seen in FIG. **4A**, power coil **90** directly contacts and operably engages with transition plates **40** of adjustable shunt systems **20** inside of induction furnace **1**. As best seen in FIG. **9**, power coil **90** may include a first end **90A**, a second end **90B** opposite to the first end **90A** (see FIG. **1**), and a lengthwise axis **90C** defined therebetween. As best seen in FIG. **9**, power coil **90** may also include an outer circumferential wall **90D** that extends between the first end **90A** and the second end **90B**. Power coil may also include an inner circumferential wall **90E** that extends between the first end **90A** and the second end **90B**. In the illustrated embodiment, inner circumferential wall **90E** is positioned interior to the outer circumferential wall **90D** and is spaced apart and separate from the outer circumferential wall **90D**. Inner circumferential wall **90E** also defines a passageway **90F** that extends between the first end **90A** and the second end **90B** for allowing a continuous stream of cooling fluid (e.g., water or other conventional cooling fluids of the like) to flow through the power coil **90** for cooling means.

[0092] Referring to FIG. **9**, power coil **90** may define an inner radius **90G** that is measured from the lengthwise axis **90C** to the inner circumferential wall **90E**. Power coil **90** may also define an outer radius **90H** that is measured from the lengthwise axis **90C** to the outer circumferential wall **90D**. With such radii **90G**, **90H**, power coil **90** may define a curvilinear and/or round cross-sectional shape along the entire length of power coil **90** between the first end **90A** and the second end **90B**. Such circular cross-sectional shape defined in power coil **90** may allow for a greater length of power coil **90** to be wound inside of induction furnace **1** as compared to conventional power coils that define a square or rectangular cross-sectional shape. Due to the implantation of the adjustable shunt systems **20** described and illustrated herein, power coil **90** may define a circular cross-sectional shape based on the structural arrangement and engagement between the power coil **90** and the transition plates **40** of adjustable shunt systems **20**.

[0093] As described in more detail below, such structural arrangement and engagement between power coil **90**, adjustable shunt systems **20**, and adjustable head system **70** enables users of induction furnace **1** to quickly remove and install a new power coil **90** than conventional methods and techniques in the art. Due to the rapid and automatic disassembly and reassembly of the induction furnace **1** via adjustable shunt systems **20** and adjustable head system **70**, power coil **90** no longer needs to be repaired inside of the induction furnace **1** as compared to conventional maintenance methods and techniques used in the art. With such ease of removing and replacing the used power coil **90** with a new power coil **90**, the used power coil **90** may be repaired offsite under optimal repair conditions.

[0094] It should be appreciated that power coil **90** is configured to connect with any suitable electrical power source that is conventionally used in the art for carrying alternating current and generating an electromagnetic field inside of induction furnace **1** to melt suitable metals. In one exemplary embodiment, power coil **90** may be formed from a high electrical conductivity oxygen free copper tubing for carrying alternating current and generating an electromagnetic field inside of induction furnace **1** to melt suitable metals.

[0095] Induction furnace **1** may also include at least quick disconnect mechanism **91** for releasably attaching or connecting at least one inlet fluid supply line “FL” to power coil **90**. As best seen in FIG. **4**, the at least one quick disconnect mechanism **91** may be operably engaged with the first end **90A** of power coil **90** to deliver cooling fluid to power coil **90**, and at least another quick disconnect mechanism (not illustrated herein) may be operably engaged with the second end **90B** of power coil **90** to return used cooling fluid to a cooling fluid source or tower (not illustrated herein) and away from the induction furnace **1**. Such use of quick disconnect mechanisms **91** is considered advantageous at least because the quick disconnect mechanisms **91** enables the induction furnace **1** to be completely isolated from a cooling fluid source and/or a main cooling fluid system. With such isolation from cooling fluid source and/or a main cooling fluid system, cooling fluid trapped inside of power coil **90** may be easily blown out and emptied when the quick

disconnect mechanisms **91** are releasably detached from the power coil **90**; such structural configuration is considered advantageous at least because power coil **90** may remain inside of the induction furnace **1** when cooling fluid is emptied from power coil **90**.

[0096] Induction furnace **1** may also include at least one cooling coil **92** that is positioned one of vertically above power coil **90** and vertically below power coil **90**. In the illustrated embodiment, induction furnace **1** may include an upper cooling coil **92A** positioned vertically above power coil **90** near the first end **90A** of power coil **90**. In this same illustrated embodiment, induction furnace **1** may also include a lower cooling coil **92B** positioned vertically below power coil **90** near the second end **90B** of power coil **90**. As such, upper cooling coil **92A** and lower cooling coil **92B** are separated and spaced apart from one another by power coil **90**. As best seen in FIGS. **1** and **9**, upper cooling coil **92A** and lower cooling coil **92B** may also define similar radii and a similar curvilinear and/or round cross-sectional shape similar to the radii and curvilinear cross-sectional shape defined by power coil **90**. In other exemplary embodiment, upper cooling coil **92A** and lower cooling coil **92B** may define radii that differ from radii **90G**, **90H** defined by power coil **90** as well as defining a cross-sectional shape that differs from the cross-sectional shape defined by power coil **90**.

[0097] Induction furnace **1** may also include at least set of quick disconnect mechanism **93** for releasably attaching or connecting at least one inlet fluid supply line “FL” to upper cooling coil **92A** and lower cooling coil **92B**. As best seen in FIG. **4**, a first set of quick disconnect mechanism **93A** may be operably engaged with the upper cooling coil **92A** to deliver cooling fluid to the upper cooling coil **92A**, and at least another quick disconnect mechanism **93B** may be operably engaged with the lower cooling coil **92B** to deliver cooling fluid to the lower cooling coil **92B**. Such use of quick disconnect mechanisms **93A**, **93B** is considered advantageous at least because the quick disconnect mechanisms **93A**, **93B** enables the induction furnace **1** to be completely isolated from a cooling fluid source and/or a main cooling fluid system. With such isolation from cooling fluid source and/or a main cooling fluid system, cooling fluid trapped inside of the upper cooling coil **92A** and the lower cooling coil **92B** may be easily blown out and emptied when the quick disconnect mechanisms **93A**, **93B** are releasably detached from the upper cooling coil **92A** and the lower cooling coil **92B**; such structural configuration is considered advantageous at least because upper cooling coil **92A** and the lower cooling coil **92B** to remain inside of the induction furnace **1** when cooling fluid is emptied from the upper cooling coil **92A** and the lower cooling coil **92B**.

[0098] Induction furnace **1** may also include a refractory liner or crucible **94** that operably engages with the power coil **90** as well as the upper cooling coil **92A** and the lower cooling coil **92B** (see FIG. **1**). Induction furnace **1** may also include grouting **96** that is positioned between the refractory liner **94** and the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B**. In other exemplary embodiments, grouting **96** may be omitted from induction furnace **1** if so desired. Induction furnace **1** may also include a bowl refractory **98** that is positioned vertically below the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** as well as the refractory liner **94**. As illustrated herein, power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** may also operably engage with bowl refractory **98** given lower cooling coil **92B** rests on the bowl refractory **98**. Bowl refractory **98** may also operably engage with the outer shell **10**, particularly, the base wall **10E** of outer shell **10**.

[0099] Induction furnace **1** may also include at least one set of load cells provided in one or both of adjustable shunt systems **20** and adjustable head system **70**. As best seen in FIGS. **1**, **5**, and **7**, induction furnace **1** may include a first set of load cells **102** that operably engages with each shunt **30** of the adjustable shunt systems **20**. In one instance, as seen in FIG. **5**, a first load cell **102A** of the first set of load cells **102** may be operably engaged with the first shunt drive assembly **24A** and the shunt **30**. Continuing with this instance, first load cell **102A** of the first set of load cells **102** may be operably engaged with the jack **28** of the first shunt drive assembly **24A** and the shunt **30**

such that the first load cell **102A** is positioned between the jack **28** of the first shunt drive assembly **24A** and the shunt **30**. In another instance, as seen in FIG. 5, a second load cell **102B** of the first set of load cells **102** may be operably engaged with the second shunt drive assembly **24B** and the shunt **30**. Continuing with this instance, second load cell **102B** of the first set of load cells **102** may be operably engaged with the jack **28** of the second shunt drive assembly **24B** and the shunt **30** such that the second load cell **102B** is positioned between the jack **28** of the second shunt drive assembly **24B** and the shunt **30**.

[0100] During operation, first set of load cells **102** may be configured to continuously monitor and measure force or pressure applied to shunt **30** by the shunt drive assemblies **24** in a first direction or by power coil **90** in a second direction. It should be appreciated that each load cell of the first set of load cells **102** may be configured to send at least one signal to a controller of induction furnace **1** when measuring force or pressure applied to shunt **30** by the shunt drive assemblies **24** in the first direction or by power coil **90** in the second direction; such controller of induction furnace **1** is described in more detail below. In one instance, each load cell of the first set of load cells **102** may be configured to periodically send signals to a controller of induction furnace **1** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**. In another instance, each load cell of the first set of load cells **102** may be configured to continuously send signals to a controller of induction furnace **1** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**.

[0101] Similarly, with respect adjustable head system **70**, induction furnace **1** may include a second set of load cells **104** that operably engages with the head **80** of the adjustable head system **70**. In one instance, as seen in FIG. 7, a first load cell **104A** of the second set of load cells **104** may be operably engaged with a head drive assembly **74** of adjustable head system **70** and the head **80**. Continuing with this instance, first load cell **104A** of the second set of load cells **104** may be operably engaged with the jack **78** of the head drive assembly **74** and the head **80** such that the first load cell **104A** is positioned between the jack **78** of the head drive assembly **74** and the head **80**.

[0102] During operation, second set of load cells **102** may also be configured to continuously monitor and measure force or pressure applied to head **80** by the head drive assemblies **74** in a first direction or by power coil **90** in a second direction. It should be appreciated that each load cell of the second set of load cells **104** may be configured to send at least one signal to a controller of induction furnace **1** when measuring force or pressure applied to head **80** by the head drive assemblies **74** in the first direction or by power coil **90** in the second direction; such controller of induction furnace **1** is described in more detail below. In one instance, each load cell of the second set of load cells **104** may be configured to periodically send signals to a controller of induction furnace **1** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**. In another instance, each load cell of the second set of load cells **104** may be configured to continuously send signals to a controller of induction furnace **1** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**.

[0103] It should be appreciated that the first set of load cells **102** described and illustrated herein may be any suitable, commercially available load cell or pressure device for continuously monitor and measure force or pressure applied to shunt **30** by the shunt drive assemblies **24** in a first direction or by power coil **90** in a second direction. It should also be appreciated that the second set of load cells **104** described and illustrated herein may be any suitable, commercially available load cell or pressure device for continuously monitor and measure force or pressure applied to head **80** by the head drive assemblies **74** in the first direction or by power coil **90** in the second direction.

[0104] Induction furnace **1** may also include a controller **110** that electrically connects with the adjustable shunt systems **20**, the adjustable head system **70**, and the sets of load cells **102**, **104**. More particularly, controller **110** electrically connects with the shunt drive assemblies **24A**, **24B** of

each adjustable shunt system **20**, head drive assemblies **74** of adjustable head system **70**, the first set of load cells **102** operably engaged with the adjustable shunt systems **20**, and the second set of load cells **104** operably engaged with the adjustable head system **70**. As best seen in FIG. **1**, a first set of electrical connections or wire **110A** electrically connects controller **110** with the shunt drive assemblies **24A**, **24B** of each adjustable shunt system **20**, a second set of electrical connections or wire **110B** electrically connects controller **110** with the head drive assemblies **74** of adjustable head system **70**, and a third set of electrical connections or wire **110C** electrically connects controller **110** with the first set of load cells **102** and the second set of load cells **104**. It should be appreciated that while first, second, and third sets of wires **110A**, **110B**, **110C** electrically connects controller **110** with the shunt drive assemblies **24A**, **24B** of each adjustable shunt system **20**, head drive assemblies **74** of adjustable head system **70**, the first set of load cells **102**, and the second set of load cells **104**, any suitable components and/or devices with suitable communication protocols may be used to electrically connect a controller with shunt drive assemblies of each adjustable shunt system, head drive assemblies of an adjustable head system, a first set of load cells operably engaged with the adjustable shunt systems, and a second set of load cells operably engaged with the adjustable head system.

[0105] During operation of induction furnace **1**, controller **110** is configured to receive information from the first set of load cells **102** as well as the second set of load cells **104** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**. Controller **110** is also configured to transmit or send signals to any one of shunt drive assemblies **24A**, **24B** of each adjustable shunt system **20** upon information received by the first set of load cells **102** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**. Similarly, controller **110** is also configured to transmit or send signals to any one of head drive assemblies **74** of adjustable head system **70** upon information received by the second set of load cells **104** during assembly of induction furnace **1**, disassembly of induction furnace **1**, or a melting process facilitated by induction furnace **1**.

[0106] Continuing with controller **110**, controller **110** may also be configured to control any one of shunt drive assemblies **24A**, **24B** of each adjustable shunt system **20** as well as any head drive assemblies **74** of adjustable head system **70**. In one instance, controller **110** may be configured to control any one of shunt drive assemblies **24A**, **24B** of each adjustable shunt system **20** between an ON state and an OFF state to automatically adjust each shunt **30** of the adjustable shunt systems **20** where each shunt **30** of the adjustable shunt systems **20** is provided at a desired pressure measured between the respective shunt drive assemblies **24A**, **24B** and the power coil **90** by measurements taken by the first set of load cells **102**. In another instance, controller **110** may be configured to control any one of head drive assemblies **74** of adjustable head system **70** between an ON state and an OFF state to automatically adjust head **80** of the adjustable head system **70** where head **80** of the adjustable head system **70** is provided at a desired pressure measured between the respective head drive assembly **74** and the power coil **90** by measurements taken by the second set of load cells **104**. It should be appreciated that controller **110** may simultaneously control each adjustable shunt system **20** and adjustable head system **70** separately and independently from one another when assembling induction furnace **1**, disassembling induction furnace **1**, or monitoring induction furnace **1** during a melting process.

[0107] It should be appreciated that any suitable, commercially available controller may be used for controller **110** to receive data and/or information from the sets of load cells **102**, **104** along with controlling the power states of the shunt drive assemblies **24** of each adjustable shunt system **20** and each head drive assembly of adjustable head system **70**. It should also be appreciated that controller **110** may be positioned at any desired location relative to the outer shell **10** of induction furnace **1**. In one instance, controller **110** may be located outside of chamber **10D** of outer shell **10** in which the controller **110** is positioned remote from induction furnace **1**. In another instance,

controller **110** may be operably engaged with the outer shell **10** and located outside of chamber **10D** of outer shell **10**. In yet another instance, controller **110** may be located inside of chamber **10D** of outer shell **10** in which the controller **110** is positioned inside of induction furnace **1**.

[0108] Having now described the systems, assemblies, and components of induction furnace **1**, methods of monitoring and adjusting induction furnace **1** during melting processes are described in greater detail below.

[0109] As seen in FIG. **10**, a metal material or charge **120** is introduced into induction furnace **1** through the central opening **82D** of the head **80**. Prior to introducing charge **120** into induction furnace **1**, an electrical current is sent to power coil **90** in order to generate an electromagnetic field inside of induction furnace **1** for either heating and/or mixing processes. Additionally, cooling fluid may also be fed to one or more of the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** at desired times during a melting process prior to introducing charge **120** into induction furnace **1**, at the time the charge **120** is introduced into induction furnace **1**, or subsequent to introducing charge **120** into induction furnace **1**. The charge **120** will then remain inside of induction furnace **1** for a desired amount of time until the charge **120** reaches a desired form (e.g., solid form, melted/liquid form, or a combination of solid and liquid form) or mixture inside of induction furnace **1**.

[0110] During this melting process, controller **110** may be configured to continuously receive information from one or both of the first set of load cells **102** and second set of load cells **104** while controlling the power states of the shunt drive assemblies **24** of each adjustable shunt system **20** and each head drive assembly of adjustable head system **70** based on the data received from the sets of load cells **102**, **104**. As best seen in FIG. **10**, at least one or both of the first load cell **102A** and the second load cell **102B** may sense or measure a pressure applied against shunt **30** during a melting process for various reasons, including force exerted outwardly by the molten charge **120**, force exerted by power coil **90**, and other various reasons that may prompt pressure to be applied against shunt **30**. Such pressure exerted against shunt **30** by at least the molten charge **120** or the power coil **90** is denoted by arrows labeled “F” in FIG. **10**.

[0111] In one instance, first load cell **102A** and second load cell **102B** may send at least one or more signals to the controller **110** during the melting process in response to pressure being applied against the shunt **30** that is within a predetermined range of pressure and/or suitable range of pressure where adjustment of the shunt **30** is not needed. In this situation, shunt drive assemblies **24A**, **24B** may remain in the OFF state in which the jacks **28** are maintained at the current position and hold the shunt **30** in place.

[0112] In another instance, first load cell **102A** and second load cell **102B** may send at least one or more signals to the controller **110** during the melting process in response to pressure being applied against the shunt **30** that is greater than a predetermined range of pressure and/or suitable range of pressure that adjustment of the shunt **30** is needed. In response to receiving such signals, controller **110** may send signals to one or both of the first shunt drive assembly **24A** and second shunt drive assembly **24B** in order to relieve and remove pressure from shunt **30**. In this illustrated embodiment, controller **110** may send a first signal to motor **26** of first shunt drive assembly **24A** to linearly move the jack **28** such that the jack **28** is withdrawn and retracted away from the power coil **90** and into the support column **22**. Such withdrawal and retraction of jack **28** in first shunt drive assembly **24A** is denoted by an arrow labeled “W1” in FIG. **10**. Given first shunt drive assembly **24A** is illustrated as a motorized jack screw mechanism, jack **28** is rotated by motor **26** in order to withdraw and retract jack **28** away from the power coil **90** and into the support column **22**. Similarly, controller **110** may send a second signal to motor **26** of second shunt drive assembly **24B** to linearly move the jack **28** such that the jack **28** is also withdrawn and retracted away from the power coil **90** and into the support column **22**. Such withdrawal and retraction of jack **28** in second shunt drive assembly **24B** is denoted by an arrow labeled “W2” in FIG. **10**. Given second shunt drive assembly **24B** is also illustrated as a motorized jack screw mechanism, jack **28** is rotated by

motor **26** in order to withdraw and retract jack **28** away from the power coil **90** and into the support column **22**. Such commands of withdrawing both jacks **28** of the first and second shunt drive assemblies **24A**, **24B** may continue until the first and second load cells **102A**, **102B** measure a pressure that is within the predetermined range of pressure to be experienced by shunt **30**.

[0113] Still referring to FIG. **10**, at least one or more load cells of the second set of load cells **104** may also sense or measure a pressure applied against head **80** during a melting process for various reasons, including force exerted outwardly by the molten charge **120**, force exerted by power coil **90**, and other various reasons that may prompt pressure to be applied against head **80**. Such pressure exerted against head **80** by at least the molten charge **120** or the power coil **90** is denoted by arrows labeled “F” in FIG. **10**.

[0114] In one instance, one or more load cell of the second set of load cells **104** may send at least one or more signals to the controller **110** during the melting process in response to pressure being applied against the head **80** that is within a predetermined range of pressure and/or suitable range of pressure where adjustment of the head **80** is not needed. In this situation, head drive assemblies **74** may remain in the OFF state in which the jacks **78** are maintained at the current position and maintain the head **80** in place.

[0115] In another instance, one or more load cell of the second set of load cells **104** may send at least one or more signals to the controller **110** during the melting process in response to the pressure being applied against the head **80** that is greater than a predetermined range of pressure and/or suitable range of pressure where adjustment of head **80** is needed. In response to such signals, controller **110** may send signals to one or more of the head drive assemblies **74** in order to relieve and remove pressure from head **80**. In this illustrated embodiment, controller **110** may send a first signal to motor **76** of a head drive assembly **74A** to linearly move the jack **78** such that the jack **78** is withdrawn and retracted away from the power coil **90** and the apron **72**. Such withdrawal and retraction of jack **78** in head drive assembly **74A** is denoted by an arrow labeled “W3” in FIG. **10**. Given head drive assembly **74A** is also illustrated as a motorized jack screw mechanism, jack **78** is rotated by motor **76** in order to withdraw and retract jack **78** away from the power coil **90** and into the apron **72**. Such commands of withdrawing jack **78** of the head drive assembly **74A** may continue until the load cell **104A** engaged with this head drive assembly **74A** measures a pressure that is within the predetermined range of pressure to be experienced by head **80**.

[0116] It should be appreciated that controller **110** may independently and separately control each first and second shunt drive assembly **24A**, **24B** in each adjustable shunt system **20** such that one or more shunts **30** of adjustable shunt systems **20** avoids experiencing extreme pressure that may lead to warpage and/or deterioration of shunts **30**. It should also be appreciated that controller **110** may independently and separately control each head drive assembly **74** in adjustable head system **70** such that head **80** of adjustable head system **70** avoids experiencing extreme pressure that may lead to warpage and/or deterioration of shunts **30**.

[0117] FIG. **11A** illustrates a set of instructions **200** that may be provided with controller **110** for monitoring each shunt **30** provided in the adjustable shunt systems **20**. Initial step **202** of set of instructions **200** may include beginning the melting process **202** inside of the induction furnace **1**. Another step **204** of set of instructions **200** may include monitoring pressure exerted on and/or against each shunt **30** provided in the adjustable shunt systems **20**. As described above, controller **110** may be assisted by first set of load cells **102** for monitoring pressure exerted on and/or against each shunt **30** provided in the adjustable shunt systems **20**. Another step **206** of set of instructions **200** may include determining if the pressure exerted on each shunt **30** provided in the adjustable shunt systems **20** is outside a range of predetermined pressures based on the measurements taken by first set of load cells **102**. If the pressure exerted on a particular shunt **30** provided in the adjustable shunt systems **20** is within the range of predetermined pressures based on the measurements taken by first set of load cells **102**, step **208** of set of instructions **200** is utilized by controller **110** in that controller **110** will have the first and second shunt drive assemblies **24A**, **24B**

maintain the position of that particular shunt **30**. If the pressure exerted on a particular shunt **30** provided in the adjustable shunt systems **20** is outside the range of predetermined pressures based on the measurements taken by first set of load cells **102**, step **210** of set of instructions **200** is utilized by controller **110** in that controller **110** will send at least one signal to first and second shunt drive assemblies **24A**, **24B** to automatically adjust the position of that particular shunt **30**.

[0118] Continuing with FIG. **11A**, another step **212** of set of instructions **200** may include determining if the pressure exerted on each shunt **30** provided in the adjustable shunt systems **20** is greater than the range of predetermined pressures based on the measurements taken by first set of load cells **102**. If the pressure exerted on a particular shunt **30** provided in the adjustable shunt systems **20** is less than the range of predetermined pressures based on the measurements taken by first set of load cells **102**, step **214** of set of instructions **200** is utilized by controller **110** in that controller **110** will have the first and second shunt drive assemblies **24A**, **24B** automatically adjust the position of that particular shunt **30** away from support column **22** and towards power coil **90**. If the pressure exerted on a particular shunt **30** provided in the adjustable shunt systems **20** is greater than the range of predetermined pressures based on the measurements taken by first set of load cells **102**, step **216** of set of instructions **200** is utilized by controller **110** in that controller **110** will send at least one signal to first and second shunt drive assemblies **24A**, **24B** to automatically adjust the position of that particular shunt **30** towards support column **22** and away from power coil **90**.

[0119] Similarly, FIG. **11B** illustrates another set of instructions **300** that may be provided with controller **110** for monitoring head **80** of the adjustable head system **70**. Initial step **302** of set of instructions **300** may include beginning the melting process **302** inside of the induction furnace **1**. Another step **304** of set of instructions **300** may include monitoring pressure exerted on and/or against head **80** of adjustable head system **70**. As described above, controller **110** may be assisted by second set of load cells **104** for monitoring pressure exerted on and/or against head **80** of adjustable head system **70**. Another step **306** of set of instructions **300** may include determining if the pressure exerted on head **80** of adjustable head systems **70** is outside a range of predetermined pressures based on the measurements taken by second set of load cells **105**. If the pressure exerted on head **80** of adjustable head system **70** is within the range of predetermined pressures based on the measurements taken by second set of load cells **104**, step **308** of set of instructions **300** is utilized by controller **110** in that controller **110** will have any one of the head drive assemblies **74** maintain the position of head **80** at a particular torque. If the pressure exerted on head **80** of adjustable head system **70** is outside the range of predetermined pressures based on the measurements taken by one of load cells of the second set of load cells **104**, step **310** of set of instructions **300** is utilized by controller **110** in that controller **110** will send at least one signal one of the head drive assemblies **74** to automatically adjust the position of head **80** at that particular location where pressure is outside the range of predetermined pressures.

[0120] Continuing with FIG. **11B**, another step **312** of set of instructions **300** may include determining if the pressure exerted on head **80** in adjustable head system **70** is greater than the range of predetermined pressures based on the measurements taken by second set of load cells **104**. If the pressure exerted on head **80** of the adjustable head system **80** is less than the range of predetermined pressures based on the measurements taken by second set of load cells **104**, step **314** of set of instructions **300** is utilized by controller **110** in that controller **110** will have at least one head drive assembly **74** automatically adjust head **80** towards apron **72** and towards power coil **90**. If the pressure exerted on head **80** of adjustable head system **70** is greater than the range of predetermined pressures based on the measurements taken by second set of load cells **104**, step **316** of set of instructions **300** is utilized by controller **110** in that controller **110** will send at least one signal to at least one head drive assembly **74** to automatically adjust the position of head **80** away from support column **22** and away from power coil **90**.

[0121] It should be appreciated that sets of instructions **200**, **300** provided with controller **110** may be performed simultaneously by controller **110** or at different time intervals by controller **100**

during a melting process. It should also be appreciated that sets of instructions **200**, **300** may be repeated in a single melting process or during multiple melting processes performed inside of induction furnace **1**.

[0122] While not illustrated herein, such monitoring of shunts **30** in each adjustable shunt system **20** and head **80** of adjustable head system **70** may be used for assembling induction furnace **1** and disassembling induction furnace **1**; such methods of assembling induction furnace **1** and disassembling induction furnace **1** with adjustable shunt system **20** and adjustable head system **70** are described in more detail below.

[0123] With respect to disassembling induction furnace **1**, adjustable shunt system **20** and adjustable head system **70**, along with assistance of the first set of load cells **102**, the second set of load cells **104**, and the controller **110**, may be performed automatically rather than using manual labor (as currently required in the art). Prior to disassembling induction furnace **1**, the shunts **30** of the adjustable shunt systems **20** and the head **80** of the adjustable head system **70** maintain the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** at a desired pressure to maintain the shape and position of these coils **90**, **92A**, **92B** inside of the induction furnace **1**. In order to remove and disassemble any one of the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B**, controller **110** may send signals to each of the first and second shunt drive assemblies **24A**, **24B** of adjustable shunt systems **20** to withdraw the shunts **30** away from the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** until the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** are free and may be removed from induction furnace **1**. Moreover, controller **110** may also send signals to each of the head drive assemblies **74** of adjustable head system **70** to withdraw the head **80** away from the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** until the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** are free and may be removed from induction furnace **1**. Upon disassembling of induction furnace **1**, head **80** may be completely removed from head drive assemblies **74** and apron **72** so that one or more of power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** may be removed from induction furnace **1**.

[0124] While disassembly of induction may be used to replace and/or repair one or more of the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B**, such disassembly of induction furnace **1** may also be performed for removing, repairing, or replacing other parts inside of induction furnace **1** discussed and illustrated herein by following the same disassembling method discussed above.

[0125] With respect to assembling induction furnace **1**, adjustable shunt system **20** and adjustable head system **70**, along with assistance of the first set of load cells **102**, the second set of load cells **104**, and the controller **110**, may be performed automatically rather than using manual labor (as currently required in the art). Initially, the outer shell **10**, adjustable shunt systems **20**, adjustable head system **70**, sets of load cells **102**, **104**, and controller **110** may be constructed and/or connected with one another prior to introduce power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B**. Once construction and connections are complete, power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** may then be introduced into induction furnace **1** and may be loosely fitted with the adjustable shunt systems **20** and adjustable head system **70**.

[0126] Once power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** are positioned inside of induction furnace **1**, the adjustable shunt systems **20** and adjustable head system **70** may begin to draw and tighten the shunts **30**, transition plates **40**, and head **80** with the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** to a desired pressure. With respect to the shunts **30** and transition plates **40**, controller **110** may send first set of signals to motors **26** of first shunt drive assemblies **24A** to linearly move the jacks **28** such that the jacks **28** are moved toward the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** and away from the support columns **22**. Similarly, controller **110** may send a second set of signals to motors **26** of second shunt drive assemblies **24B** to linearly move the jacks **28** such that the jacks **28** are moved toward

the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** and away from the support columns **22**. The controller **110** may continue to control the first and second shunt drive assemblies **24A**, **24B** to press the shunts **30** and transition plates **40** towards the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** based on the pressure measurements taken by the first set of load cells **102**. As such, the controller **110** may continue to control the first and second shunt drive assemblies **24A**, **24B** to press the shunts **30** towards the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** until a desired pressure is measured by the first set of load cells **102**. At this point, the shunts **30** and transition plates **40** of the adjustable shunt system **20** are engaged with the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** at a desired torque and/or pressure exerted and maintained by first and second shunt drive assemblies **24A**, **24B**.

[0127] With respect to the head **80**, controller **110** may send a third set of signals to motors **76** of head drive assemblies **74** to linearly move the jacks **78** such that the head **80** is drawn downwardly towards the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** and towards the apron **72**. The controller **110** may continue to control the head drive assemblies **74** to draw the head **80** downwardly on the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** based on the pressure measurements taken by the second set of load cells **104**. As such, the controller **110** may continue to control the head drive assemblies **74** to press the head **80** against the power coil **90**, upper cooling coil **92A**, and lower cooling coil **92B** until a desired pressure is measured by the second set of load cells **104**. At this point, the head **80** of the adjustable head system **70** is engaged with the power coil **90**, the upper cooling coil **92A**, and the lower cooling coil **92B** at a desired torque and/or pressure exerted and maintained by head drive assemblies **74**.

[0128] FIG. **12A** illustrates a set of instructions **400** that may be provided with controller **110** for assembling each shunt **30** provided in the adjustable shunt systems **20** with at least power coil **90** at a desired torque. Initial step **402** of set of instructions **400** may include beginning the assembling process of the induction furnace **1**. Another step **404** of set of instructions **400** may include controller **110** sending at least one signal to first and second shunt drive assemblies **24A**, **24B** to automatically adjust the position of shunt **30** of each adjustable shunt system **20**. Another step **406** of set of instructions **400** may include monitoring pressure exerted on and/or against each shunt **30** provided in the adjustable shunt systems **20**. As described above, controller **110** may be assisted by first set of load cells **102** for monitoring pressure exerted on and/or against each shunt **30** provide in the adjustable shunt systems **20**. Another step **408** of set of instructions **400** may include determining if the desired torque is applied on a particular shunt **30** and power coil **90** based on the measurements taken by first set of load cells **102**. If the desired torque between a particular shunt **30** and power coil **90** is not met based on the measurements taken by first set of load cells **102**, step **410** of set of instructions **400** is utilized by controller **110** in that controller **110** continues to automatically adjust this particular shunt **30** until the desired torque is measured between this particular shunt **30** and power coil **90**; as such, steps **404** and **406** will be repeated until the desired torque is measured between this particular shunt **30** and power coil **90**. If the desired torque between a particular shunt **30** and power coil **90** is met based on the measurements taken by first set of load cells **102**, step **412** of set of instructions **400** is utilized by controller **110** in that controller **110** will have the first and second shunt drive assemblies **24A**, **24B** maintain the position of that particular shunt **30**.

[0129] Similarly, FIG. **12B** illustrates a set of instructions **500** that may be provided with controller **110** for assembling head **80** of adjustable head system **70** with at least power coil **90** at a desired torque. Initial step **502** of set of instructions **500** may include beginning the assembling process of the induction furnace **1**. Another step **504** of set of instructions **500** may include controller **110** sending at least one signal to head drive assemblies **74** to automatically adjust the position of head **80** of adjustable head system **70**. Another step **506** of set of instructions **500** may include monitoring pressure exerted on and/or against head **80** of adjustable head system **70**. As described above, controller **110** may be assisted by second set of load cells **104** for monitoring pressure

exerted on and/or against head **80** of adjustable head system **70**. Another step **508** of set of instructions **500** may include determining if the desired torque is applied at specific locations on head **80** and power coil **90** based on the measurements taken by second set of load cells **102**. If the desired torque between head **80** and power coil **90** is not met based on the measurements taken by one load cell of the second set of load cells **104**, step **510** of set of instructions **500** is utilized by controller **110** in that controller **110** continues to automatically adjust head **80** at this particular location until the desired torque is measured between head **80** and power coil **90**; as such, steps **504** and **506** will be repeated until the desired torque is measured between head **80** and power coil **90**. If the desired torque between head **80** and power coil **90** is met based on the measurements taken by second set of load cells **104**, step **512** of set of instructions **500** is utilized by controller **110** in that controller **110** will have the head drive assemblies **74** maintain the position of head **80** with the power coil **90**.

[0130] It should be appreciated that sets of instructions **400**, **500** provided with controller **110** may be performed simultaneously by controller **110** or at different time intervals by controller **100** during assembly or disassembly of induction furnace **1**. It should also be appreciated that sets of instructions **400**, **500** may be repeated in a single assembly or disassembly process or during multiple assembly or disassembly processes of induction furnace **1**.

[0131] FIG. **13** illustrates a method **600** of automatically adjusting a shunt of a shunt system of an induction furnace. An initial step **602** of method **600** may include engaging at least one shunt drive assembly of the shunt system with a support column of the shunt system. Another step **604** of method **600** may include engaging the at least one shunt drive assembly with the shunt of the shunt system. Another step **606** of method **600** may include engaging at least one cooling coil of the induction furnace and a power coil of the induction furnace with the shunt. Another step **608** of method **600** may include automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least one shunt drive assembly, relative to the at least one cooling coil and the power coil.

[0132] In other exemplary embodiments, method **600** may include optional or additional steps of automatically adjusting a shunt of a shunt system of an induction furnace. An optional step may further include advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. An optional step may further include withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least one load cell with the at least one shunt drive assembly and the shunt; and measuring pressure at at least one position on the shunt by the at least one load cell. Optional steps of may further include sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a predetermined pressure applied against the shunt; and advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a predetermined pressure applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least another drive assembly with the shunt of the shunt system; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least another drive assembly, relative to the at least one cooling coil and the power coil; wherein the at least one shunt drive assembly and the at least

another drive assembly separately and independently adjust the shunt.

[0133] FIG. **14** illustrates a method **700** of automatically adjusting a head of a head system of an induction furnace. An initial step **702** of method **700** may include engaging at least one head drive assembly of the head system with an apron of the head system. Another step **704** of method **700** may include engaging the at least one head drive assembly with the head of the head system. Another step **706** of method **700** may include engaging at least one cooling coil of the induction furnace with the head. Another step **708** of method **700** may include automatically adjusting the head of the head system of the induction furnace, via the at least one head drive assembly, relative to the at least one cooling coil.

[0134] In other exemplary embodiments, method **700** may include optional or additional steps of automatically adjusting a head of a head system of an induction furnace. An optional step may further include retracting the head towards the apron and towards the at least one cooling coil and a power coil of the induction furnace, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. An optional step may further include withdrawing the head away from the at least one cooling coil and a power coil of the induction furnace and towards the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least one load cell with the at least one head drive assembly and the head, and measuring pressure at least one position on the head by the at least one load cell. Optional steps may further include sending at least one signal to a controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and retracting the head towards the apron and towards the at least one cooling coil and a power coil of the induction furnace, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from the at least one cooling coil and a power coil of the induction furnace and towards the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least another head drive assembly with the head of the head system; and automatically adjusting the head of the head system of the induction furnace, via the at least another head drive assembly, relative to the at least one cooling coil and a power coil of the induction furnace; wherein the at least one head drive assembly and the at least another head drive assembly separately and independently adjust the head.

[0135] FIG. **15** illustrates a method **800** of monitoring pressure applied to a shunt of an induction furnace during a melting process. An initial step **802** of method **800** may include engaging at least one shunt drive assembly of a shunt system with a support column of the shunt system and the shunt of the shunt system. Another step **804** of method **800** may include providing at least one load cell with the shunt at a first position and with the at least one shunt drive assembly of the shunt system. Another step **806** of method **800** may include connecting a controller of the induction furnace with the at least one load cell. Another step **808** of method **800** may include connecting the controller with the at least one shunt drive assembly. Another step **810** of method **800** may include monitoring pressure applied to the shunt of the induction furnace during the melting process by the at least one load cell.

[0136] Optional steps and/or additional steps may further include with method **800** discussed above. An optional step may further include automatically adjusting the shunt of the shunt system, via the at least one shunt drive assembly of the shunt system, relative to at least one cooling coil of the induction furnace and a power coil of the induction furnace during the melting process. Optional steps may further include sending at least one signal to the controller, via the at least one

load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards at least one cooling coil of the induction furnace and the power coil of the induction furnace, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil of the induction furnace and the power coil of the induction furnace and towards the support column, via the at least one shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least another shunt drive assembly of the shunt system with the shunt of the shunt system; providing at least another load cell with the shunt at a second position and with the at least another shunt drive assembly; and monitoring pressure applied to the shunt by the at least another load cell; wherein the second position is different than the first position. Method may further include that the at least one shunt drive assembly and the at least another shunt drive assembly separately and independently automatically adjust the shunt; and the at least one load cell and the at least another load cell of the shunt system separately and independently monitor the shunt. Optional steps may further include sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the least another cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards at least one cooling coil and a power coil, via at least one of the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via one of the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the shunt; and advancing the shunt away from the support column and towards the at least one cooling coil and the power coil, via the at least one shunt drive assembly and the at least another shunt drive assembly of the shunt system, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the shunt; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the shunt; and withdrawing the shunt away from the at least one cooling coil and the power coil and towards the support column, via the at least one shunt drive assembly and the at least another shunt drive assembly, until a desired pressure is maintained between the shunt, the at least one cooling coil, and the power coil.

[0137] FIG. 16 illustrates a method 900 of monitoring pressure applied to a head of an induction furnace during a melting process. An initial step 902 of method 900 may include engaging at least one head drive assembly of a head system with an apron of the head system and the head of the

head system. Another step **904** of method **900** may include providing at least one load cell with the head at a first position and with the at least one head drive assembly of the head system. Another step **906** of method **900** may include connecting a controller of the induction furnace with the at least one load cell. Another step **908** of method **900** may include connecting the controller with the at least one head drive assembly. Another step **910** of method **900** may include monitoring pressure applied to the head during the melting process by the at least one load cell.

[0138] Optional steps and/or additional steps may further include with method **900** discussed above. An optional step may further include automatically adjusting the head of the head system, via the at least one head drive assembly, relative to at least one cooling coil of the induction furnace or a power coil of the induction furnace during the melting process. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and retracting the head towards the apron and towards at least one cooling coil and a power coil, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell of the head system measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from at least one cooling coil and a power coil and away from the apron, via the at least one head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include engaging at least another head drive assembly with the head and the apron; providing at least another load cell with the head at a second position and with the at least another head drive assembly; and monitoring pressure applied to the head by the at least another load cell; wherein the second position is different than the first position. Optional steps may further include sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; and advancing the head towards the apron and towards at least one cooling coil and a power coil, via at least one of the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell or the at least another load cell, when the at least one load cell or the at least another load cell measures a pressure that is outside a range of predetermined pressures applied against the head; and withdrawing the head away from the apron and away from the at least one cooling coil and the power coil, via at least one of the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the head; and advancing the head towards the apron and towards the at least one cooling coil and the power coil, via the at least one head drive assembly and the at least another head drive assembly, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil. Optional steps may further include sending at least one signal to the controller, via the at least one load cell, when the at least one load cell measures a first pressure that is outside a range of predetermined pressures applied against the head; sending at least another signal to the controller, via the at least another load cell, when the at least another load cell measures a second pressure that is outside the range of predetermined pressures applied against the head; and withdrawing the head away from the apron and away from the at least one cooling coil and the power coil, via at least one

of the at least one head drive assembly and the at least another head drive assembly of the head system, until a desired pressure is maintained between the head, the at least one cooling coil, and the power coil.

[0139] FIG. **17** illustrates a method **1000** of monitoring pressure applied to a shunt and a head of an induction furnace during a melting process. An initial step **1002** of method **1000** may include engaging at least one shunt drive assembly of a shunt system with a support column of the shunt system and the shunt of the shunt system. Another step **1004** of method **1000** may include engaging at least one head drive assembly of a head system with an apron of the head system and the head of the head system. Another step **1006** of method **1000** may include engaging at least one cooling coil of the induction furnace and a power coil of the induction furnace with the shunt. Another step **1008** of method **1000** may include engaging the at least one cooling coil of the induction furnace with the head. Another step **1010** of method **1000** may include providing at least one load cell with the shunt and with the at least one shunt drive assembly of the shunt system. Another step of **1012** of method **1000** may include providing at least another load cell with the head and with the at least one head drive assembly of the head system. Another step **1014** of method **1000** may include connecting a controller of the induction furnace with the at least one load cell and the at least another load cell. Another step **1016** of method **1000** may include connecting the controller with the at least one shunt drive assembly and the at least one head drive assembly. Another step **1018** of method **1000** may include monitoring pressure applied to the shunt and the head of the induction furnace during the melting process.

[0140] FIG. **18** illustrates a method **1100** of automatically adjusting one of a shunt of a shunt system and a head of a head system of a coreless induction furnace. An initial step **1102** of method **1100** may include engaging the shunt system with an outer shell of the coreless induction furnace. Another step **1104** of method **1100** may include engaging the head system with the outer shell of the coreless induction furnace. Another step **1106** of method **1100** may include engaging at least one cooling coil of the coreless induction furnace with the shunt system and the head system. Another step **1108** of method **1100** may include engaging a power coil of the coreless induction furnace with the shunt system and the at least one cooling coil. Another step **1110** of method **1100** may include automatically adjusting one of the shunt of the shunt system and a head of the head system of the coreless induction furnace.

[0141] Optional steps and/or additional steps may further include with method **1100** discussed above. Optional steps may further include that the steps of engaging the shunt system with the outer shell, the at least one cooling coil, and the power coil further comprises: engaging a support column of the shunt system with the outer shell; engaging at least one shunt drive assembly of the shunt system with the support column of the shunt system; engaging the at least one shunt drive assembly with the shunt of the shunt system; engaging the at least one cooling coil of the coreless induction furnace and the power coil of the coreless induction furnace with the shunt; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least one shunt drive assembly, relative to the at least one cooling coil and the power coil. Optional steps may further include that the steps of engaging the head system with the outer shell and the at least one cooling coil further comprises: engaging an apron of the shunt system with the outer shell; engaging at least one head drive assembly of the head system with the apron of the head system; engaging the at least one head drive assembly with the head of the head system; engaging at least one cooling coil of the coreless induction furnace with the head; and automatically adjusting the head of the head system of the coreless induction furnace, via the at least one head drive assembly, relative to the at least one cooling coil.

[0142] The device, assembly, or system of the present disclosure may additionally include one or more sensor to sense or gather data pertaining to the surrounding environment or operation of the device, assembly, or system. Some exemplary sensors capable of being electronically coupled with the device, assembly, or system of the present disclosure (either directly connected to the device,

assembly, or system of the present disclosure or remotely connected thereto) may include but are not limited to: accelerometers sensing accelerations experienced during rotation, translation, velocity/speed, location traveled, elevation gained; gyroscopes sensing movements during angular orientation and/or rotation, and rotation; altimeters sensing barometric pressure, altitude change, terrain climbed, local pressure changes, submersion in liquid; impellers measuring the amount of fluid passing thereby; Global Positioning sensors sensing location, elevation, distance traveled, velocity/speed; audio sensors sensing local environmental sound levels, or voice detection; Photo/Light sensors sensing ambient light intensity, ambient, Day/night, UV exposure; TV/IR sensors sensing light wavelength; Temperature sensors sensing machine or motor temperature, ambient air temperature, and environmental temperature; and Moisture Sensors sensing surrounding moisture levels.

[0143] The device, assembly, or system of the present disclosure may include wireless communication logic coupled to sensors on the device, assembly, or system. The sensors gather data and provide the data to the wireless communication logic. Then, the wireless communication logic may transmit the data gathered from the sensors to a remote device. Thus, the wireless communication logic may be part of a broader communication system, in which one or several devices, assemblies, or systems of the present disclosure may be networked together to report alerts and, more generally, to be accessed and controlled remotely. Depending on the types of transceivers installed in the device, assembly, or system of the present disclosure, the system may use a variety of protocols (e.g., Wifi, ZigBee, MiWi, Bluetooth) for communication. In one example, each of the devices, assemblies, or systems of the present disclosure may have its own IP address and may communicate directly with a router or gateway. This would typically be the case if the communication protocol is WiFi.

[0144] In another example, a point-to-point communication protocol like MiWi or ZigBee is used. One or more of the device, assembly, or system of the present disclosure may serve as a repeater, or the devices, assemblies, or systems of the present disclosure may be connected together in a mesh network to relay signals from one device, assembly, or system to the next. However, the individual device, assembly, or system in this scheme typically would not have IP addresses of their own. Instead, one or more of the devices, assemblies, or system of the present disclosure communicates with a repeater that does have an IP address, or another type of address, identifier, or credential needed to communicate with an outside network. The repeater communicates with the router or gateway.

[0145] In either communication scheme, the router or gateway communicates with a communication network, such as the Internet, although in some embodiments, the communication network may be a private network that uses transmission control protocol/internet protocol (TCP/IP) and other common Internet protocols but does not interface with the broader Internet, or does so only selectively through a firewall.

[0146] The system that receives and processes signals from the device, assembly, or system of the present disclosure may differ from embodiment to embodiment. In one embodiment, alerts and signals from the device, assembly, or system of the present disclosure are sent through an e-mail or simple message service (SMS; text message) gateway so that they can be sent as e-mails or SMS text messages to a remote device, such as a smartphone, laptop, or tablet computer, monitored by a responsible individual, group of individuals, or department, such as a maintenance department. Thus, if a particular device, assembly, or system of the present disclosure creates an alert because of a data point gathered by one or more sensors, that alert can be sent, in e-mail or SMS form, directly to the individual responsible for fixing it. Of course, e-mail and SMS are only two examples of communication methods that may be used; in other embodiments, different forms of communication may be used.

[0147] In other embodiments, alerts and other data from the sensors on the device, assembly, or system of the present disclosure may also be sent to a work tracking system that allows the

individual, or the organization for which he or she works, to track the status of the various alerts that are received, to schedule particular workers to repair a particular device, assembly, or system of the present disclosure, and to track the status of those repair jobs. A work tracking system would typically be a server, such as a Web server, which provides an interface individuals and organizations can use, typically through the communication network. In addition to its work tracking functions, the work tracker may allow broader data logging and analysis functions. For example, operational data may be calculated from the data collected by the sensors on the device, assembly, or system of the present disclosure, and the system may be able to provide aggregate machine operational data for a device, assembly, or system of the present disclosure or group of devices, assemblies, or systems of the present disclosure.

[0148] The system also allows individuals to access the device, assembly, or system of the present disclosure for configuration and diagnostic purposes. In that case, the individual processors or microcontrollers of the device, assembly, or system of the present disclosure may be configured to act as Web servers that use a protocol like hypertext transfer protocol (HTTP) to provide an online interface that can be used to configure the device, assembly, or system. In some embodiments, the systems may be used to configure several devices, assemblies, or systems of the present disclosure at once. For example, if several devices, assemblies, or systems are of the same model and are in similar locations in the same location, it may not be necessary to configure the devices, assemblies, or systems individually. Instead, an individual may provide configuration information, including baseline operational parameters, for several devices, assemblies, or systems at once.

[0149] As described herein, aspects of the present disclosure may include one or more electrical, pneumatic, hydraulic, or other similar secondary components and/or systems therein. The present disclosure is therefore contemplated and will be understood to include any necessary operational components thereof. For example, electrical components will be understood to include any suitable and necessary wiring, fuses, or the like for normal operation thereof. Similarly, any pneumatic systems provided may include any secondary or peripheral components such as air hoses, compressors, valves, meters, or the like. It will be further understood that any connections between various components not explicitly described herein may be made through any suitable means including mechanical fasteners, or more permanent attachment means, such as welding or the like. Alternatively, where feasible and/or desirable, various components of the present disclosure may be integrally formed as a single unit.

[0150] Various inventive concepts may be embodied as one or more methods, of which an example has been provided. The acts performed as part of the method may be ordered in any suitable way. Accordingly, embodiments may be constructed in which acts are performed in an order different than illustrated, which may include performing some acts simultaneously, even though shown as sequential acts in illustrative embodiments.

[0151] While various inventive embodiments have been described and illustrated herein, those of ordinary skill in the art will readily envision a variety of other means and/or structures for performing the function and/or obtaining the results and/or one or more of the advantages described herein, and each of such variations and/or modifications is deemed to be within the scope of the inventive embodiments described herein. More generally, those skilled in the art will readily appreciate that all parameters, dimensions, materials, and configurations described herein are meant to be exemplary and that the actual parameters, dimensions, materials, and/or configurations will depend upon the specific application or applications for which the inventive teachings is/are used. Those skilled in the art will recognize, or be able to ascertain using no more than routine experimentation, many equivalents to the specific inventive embodiments described herein. It is, therefore, to be understood that the foregoing embodiments are presented by way of example only and that, within the scope of the appended claims and equivalents thereto, inventive embodiments may be practiced otherwise than as specifically described and claimed. Inventive embodiments of the present disclosure are directed to each individual feature, system, article, material, kit, and/or

method described herein. In addition, any combination of two or more such features, systems, articles, materials, kits, and/or methods, if such features, systems, articles, materials, kits, and/or methods are not mutually inconsistent, is included within the inventive scope of the present disclosure.

[0152] The above-described embodiments can be implemented in any of numerous ways. For example, embodiments of technology disclosed herein may be implemented using hardware, software, or a combination thereof. When implemented in software, the software code or instructions can be executed on any suitable processor or collection of processors, whether provided in a single computer or distributed among multiple computers. Furthermore, the instructions or software code can be stored in at least one non-transitory computer readable storage medium.

[0153] Also, a computer or smartphone may be utilized to execute the software code or instructions via its processors may have one or more input and output devices. These devices can be used, among other things, to present a user interface. Examples of output devices that can be used to provide a user interface include printers or display screens for visual presentation of output and speakers or other sound generating devices for audible presentation of output. Examples of input devices that can be used for a user interface include keyboards, and pointing devices, such as mice, touch pads, and digitizing tablets. As another example, a computer may receive input information through speech recognition or in other audible format.

[0154] Such computers or smartphones may be interconnected by one or more networks in any suitable form, including a local area network or a wide area network, such as an enterprise network, and intelligent network (IN) or the Internet. Such networks may be based on any suitable technology and may operate according to any suitable protocol and may include wireless networks, wired networks or fiber optic networks.

[0155] The various methods or processes outlined herein may be coded as software/instructions that is executable on one or more processors that employ any one of a variety of operating systems or platforms. Additionally, such software may be written using any of a number of suitable programming languages and/or programming or scripting tools, and also may be compiled as executable machine language code or intermediate code that is executed on a framework or virtual machine.

[0156] In this respect, various inventive concepts may be embodied as a computer readable storage medium (or multiple computer readable storage media) (e.g., a computer memory, one or more floppy discs, compact discs, optical discs, magnetic tapes, flash memories, USB flash drives, SD cards, circuit configurations in Field Programmable Gate Arrays or other semiconductor devices, or other non-transitory medium or tangible computer storage medium) encoded with one or more programs that, when executed on one or more computers or other processors, perform methods that implement the various embodiments of the disclosure discussed above. The computer readable medium or media can be transportable, such that the program or programs stored thereon can be loaded onto one or more different computers or other processors to implement various aspects of the present disclosure as discussed above.

[0157] The terms “program” or “software” or “instructions” are used herein in a generic sense to refer to any type of computer code or set of computer-executable instructions that can be employed to program a computer or other processor to implement various aspects of embodiments as discussed above. Additionally, it should be appreciated that according to one aspect, one or more computer programs that when executed perform methods of the present disclosure need not reside on a single computer or processor, but may be distributed in a modular fashion amongst a number of different computers or processors to implement various aspects of the present disclosure.

[0158] Computer-executable instructions may be in many forms, such as program modules, executed by one or more computers or other devices. Generally, program modules include routines, programs, objects, components, data structures, etc. that perform particular tasks or implement

particular abstract data types. Typically the functionality of the program modules may be combined or distributed as desired in various embodiments.

[0159] Also, data structures may be stored in computer-readable media in any suitable form. For simplicity of illustration, data structures may be shown to have fields that are related through location in the data structure. Such relationships may likewise be achieved by assigning storage for the fields with locations in a computer-readable medium that convey relationship between the fields. However, any suitable mechanism may be used to establish a relationship between information in fields of a data structure, including through the use of pointers, tags or other mechanisms that establish relationship between data elements.

[0160] All definitions, as defined and used herein, should be understood to control over dictionary definitions, definitions in documents incorporated by reference, and/or ordinary meanings of the defined terms.

[0161] “Logic”, as used herein, includes but is not limited to hardware, firmware, software, and/or combinations of each to perform a function(s) or an action(s), and/or to cause a function or action from another logic, method, and/or system. For example, based on a desired application or needs, logic may include a software controlled microprocessor, discrete logic like a processor (e.g., microprocessor), an application specific integrated circuit (ASIC), a programmed logic device, a memory device containing instructions, an electric device having a memory, or the like. Logic may include one or more gates, combinations of gates, or other circuit components. Logic may also be fully embodied as software. Where multiple logics are described, it may be possible to incorporate the multiple logics into one physical logic. Similarly, where a single logic is described, it may be possible to distribute that single logic between multiple physical logics.

[0162] Furthermore, the logic(s) presented herein for accomplishing various methods of this system may be directed towards improvements in existing computer-centric or internet-centric technology that may not have previous analog versions. The logic(s) may provide specific functionality directly related to structure that addresses and resolves some problems identified herein. The logic(s) may also provide significantly more advantages to solve these problems by providing an exemplary inventive concept as specific logic structure and concordant functionality of the method and system. Furthermore, the logic(s) may also provide specific computer implemented rules that improve on existing technological processes. The logic(s) provided herein extends beyond merely gathering data, analyzing the information, and displaying the results. Further, portions or all of the present disclosure may rely on underlying equations that are derived from the specific arrangement of the equipment or components as recited herein. Thus, portions of the present disclosure as it relates to the specific arrangement of the components are not directed to abstract ideas.

Furthermore, the present disclosure and the appended claims present teachings that involve more than performance of well-understood, routine, and conventional activities previously known to the industry. In some of the method or process of the present disclosure, which may incorporate some aspects of natural phenomenon, the process or method steps are additional features that are new and useful.

[0163] The articles “a” and “an,” as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean “at least one.” The phrase “and/or,” as used herein in the specification and in the claims (if at all), should be understood to mean “either or both” of the elements so conjoined, i.e., elements that are conjunctively present in some cases and disjunctively present in other cases. Multiple elements listed with “and/or” should be construed in the same fashion, i.e., “one or more” of the elements so conjoined. Other elements may optionally be present other than the elements specifically identified by the “and/or” clause, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, a reference to “A and/or B”, when used in conjunction with open-ended language such as “comprising” can refer, in one embodiment, to A only (optionally including elements other than B); in another embodiment, to B only (optionally including elements other than A); in yet another

embodiment, to both A and B (optionally including other elements); etc. As used herein in the specification and in the claims, “or” should be understood to have the same meaning as “and/or” as defined above. For example, when separating items in a list, “or” or “and/or” shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of elements, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as “only one of” or “exactly one of,” or, when used in the claims, “consisting of,” will refer to the inclusion of exactly one element of a number or list of elements. In general, the term “or” as used herein shall only be interpreted as indicating exclusive alternatives (i.e. “one or the other but not both”) when preceded by terms of exclusivity, such as “either,” “one of,” “only one of,” or “exactly one of.” “Consisting essentially of,” when used in the claims, shall have its ordinary meaning as used in the field of patent law.

[0164] As used herein in the specification and in the claims, the phrase “at least one,” in reference to a list of one or more elements, should be understood to mean at least one element selected from any one or more of the elements in the list of elements, but not necessarily including at least one of each and every element specifically listed within the list of elements and not excluding any combinations of elements in the list of elements. This definition also allows that elements may optionally be present other than the elements specifically identified within the list of elements to which the phrase “at least one” refers, whether related or unrelated to those elements specifically identified. Thus, as a non-limiting example, “at least one of A and B” (or, equivalently, “at least one of A or B,” or, equivalently “at least one of A and/or B”) can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including elements other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including elements other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other elements); etc.

[0165] As used herein in the specification and in the claims, the term “effecting” or a phrase or claim element beginning with the term “effecting” should be understood to mean to cause something to happen or to bring something about. For example, effecting an event to occur may be caused by actions of a first party even though a second party actually performed the event or had the event occur to the second party. Stated otherwise, effecting refers to one party giving another party the tools, objects, or resources to cause an event to occur. Thus, in this example a claim element of “effecting an event to occur” would mean that a first party is giving a second party the tools or resources needed for the second party to perform the event, however the affirmative single action is the responsibility of the first party to provide the tools or resources to cause said event to occur.

[0166] When a feature or element is herein referred to as being “on” another feature or element, it can be directly on the other feature or element or intervening features and/or elements may also be present. In contrast, when a feature or element is referred to as being “directly on” another feature or element, there are no intervening features or elements present. It will also be understood that, when a feature or element is referred to as being “connected”, “attached” or “coupled” to another feature or element, it can be directly connected, attached or coupled to the other feature or element or intervening features or elements may be present. In contrast, when a feature or element is referred to as being “directly connected”, “directly attached” or “directly coupled” to another feature or element, there are no intervening features or elements present. Although described or shown with respect to one embodiment, the features and elements so described or shown can apply to other embodiments. It will also be appreciated by those of skill in the art that references to a structure or feature that is disposed “adjacent” another feature may have portions that overlap or underlie the adjacent feature.

[0167] Spatially relative terms, such as “under”, “below”, “lower”, “over”, “upper”, “above”, “behind”, “in front of”, and the like, may be used herein for ease of description to describe one

element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. It will be understood that the spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. For example, if a device in the figures is inverted, elements described as “under” or “beneath” other elements or features would then be oriented “over” the other elements or features. Thus, the exemplary term “under” can encompass both an orientation of over and under. The device may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein interpreted accordingly. Similarly, the terms “upwardly”, “downwardly”, “vertical”, “horizontal”, “lateral”, “transverse”, “longitudinal”, and the like are used herein for the purpose of explanation only unless specifically indicated otherwise.

[0168] Although the terms “first” and “second” may be used herein to describe various features/elements, these features/elements should not be limited by these terms, unless the context indicates otherwise. These terms may be used to distinguish one feature/element from another feature/element. Thus, a first feature/element discussed herein could be termed a second feature/element, and similarly, a second feature/element discussed herein could be termed a first feature/element without departing from the teachings of the present invention.

[0169] An embodiment is an implementation or example of the present disclosure. Reference in the specification to “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, means that a particular feature, structure, or characteristic described in connection with the embodiments is included in at least some embodiments, but not necessarily all embodiments, of the invention. The various appearances “an embodiment,” “one embodiment,” “some embodiments,” “one particular embodiment,” “an exemplary embodiment,” or “other embodiments,” or the like, are not necessarily all referring to the same embodiments. Furthermore, the use of any and all examples or exemplary language (“e.g.,” “such as,” or the like) is intended merely to better illustrate or illuminate the embodiments and does not pose a limitation on the scope of that or those embodiments. No language in this specification should be construed as indicating any unclaimed element as essential to the practice of the disclosed embodiment.

[0170] If this specification states a component, feature, structure, or characteristic “may”, “might”, or “could” be included, that particular component, feature, structure, or characteristic is not required to be included. If the specification or claim refers to “a” or “an” element, that does not mean there is only one of the element. If the specification or claims refer to “an additional” element, that does not preclude there being more than one of the additional element.

[0171] As used herein in the specification and claims, including as used in the examples and unless otherwise expressly specified, all numbers may be read as if prefaced by the word “about” or “approximately,” even if the term does not expressly appear. The phrase “about” or “approximately” may be used when describing magnitude and/or position to indicate that the value and/or position described is within a reasonable expected range of values and/or positions. For example, a numeric value may have a value that is $\pm 0.1\%$ of the stated value (or range of values), $\pm 1\%$ of the stated value (or range of values), $\pm 2\%$ of the stated value (or range of values), $\pm 5\%$ of the stated value (or range of values), $\pm 10\%$ of the stated value (or range of values), etc. Any numerical range recited herein is intended to include all sub-ranges subsumed therein.

[0172] Additionally, the method of performing the present disclosure may occur in a sequence different than those described herein. Accordingly, no sequence of the method should be read as a limitation unless explicitly stated. It is recognizable that performing some of the steps of the method in a different order could achieve a similar result.

[0173] In the claims, as well as in the specification above, all transitional phrases such as “comprising,” “including,” “carrying,” “having,” “containing,” “involving,” “holding,” “composed of,” and the like are to be understood to be open-ended, i.e., to mean including but not limited to. Only the transitional phrases “consisting of” and “consisting essentially of” shall be closed or semi-

closed transitional phrases, respectively.

[0174] To the extent that the present disclosure has utilized the term “invention” in various titles or sections of this specification, or in the context of those sections, this term has been included as required by the formatting requirements of word document submissions (i.e., docx submissions) pursuant the guidelines/requirements of the United States Patent and Trademark Office and shall not, in any manner, be considered a disavowal of any subject matter.

[0175] In the foregoing description, certain terms have been used for brevity, clearness, and understanding. No unnecessary limitations are to be implied therefrom beyond the requirement of the prior art because such terms are used for descriptive purposes and are intended to be broadly construed.

[0176] Moreover, the description and illustration of various embodiments of the disclosure are examples and the disclosure is not limited to the exact details shown or described.

Claims

1. A coreless induction furnace, comprising: an outer shell; a set of adjustable shunt systems operably engaged with the outer shell; at least one cooling coil operably engaged with the set of adjustable shunt systems; and a power coil operably engaged with the at least one cooling coil and the set of adjustable shunt systems; wherein the set of adjustable shunt systems is configured to automatically adjust at least one shunt of the set of adjustable shunt systems relative to at least one of the at least one cooling coil and the power coil.
2. The coreless induction furnace of claim 1, wherein each adjustable shunt system of the adjustable shunt systems comprises: a support column operably engaged with the outer shell; and at least one shunt drive assembly operably engaged with the support column and the at least one shunt; wherein the shunt operably engages with the at least one cooling coil and the power coil; and wherein the at least one shunt drive assembly is configured to automatically adjust the shunt along an axis relative to the support column.
3. The coreless induction furnace of claim 2, wherein when the shunt is adjusted by the at least one shunt drive assembly, the at least one cooling coil and the power coil are adjusted with the shunt.
4. The coreless induction furnace of claim 2, wherein the at least one shunt drive assembly comprises: a first jack having a first end that operably engages with the shunt and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the shunt along the axis that is orthogonal to the support column.
5. The coreless induction furnace of claim 2, wherein each adjustable shunt system of the set of adjustable shunt systems further comprises: at least another shunt drive assembly operably engaged with the support column and the shunt; wherein the at least another shunt drive assembly is configured to automatically adjust the shunt along a second axis that is orthogonal to the support column and parallel with the axis of the at least one shunt drive assembly; and wherein each of the at least one shunt drive assembly and the at least another shunt drive assembly adjusts the shunt independently and separately from one another.
6. The coreless induction furnace of claim 2, wherein each adjustable shunt system of the set of adjustable shunt systems further comprises: a transition plate operably engaged with the shunt, the at least one cooling coil, and the power coil; wherein the transition plate is positioned between the shunt and the at least one cooling coil and the power coil such that the shunt is spaced apart from the at least one cooling coil and the power coil.
7. The coreless induction furnace of claim 2, wherein each adjustable shunt system of the set of adjustable shunt systems further comprises: a track operably engaged with the support column; and a sled assembly operably engaged with the track and operably engaged with the shunt at one of a first end of the shunt and a second end of the shunt opposite to the first end; wherein the sled

assembly is configured to guide the shunt along the track.

8. The coreless induction furnace of claim 2, further comprising: at least one load cell operably engaged with the shunt and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; and a controller electrically connected with the at least one load cell and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; wherein the controller is configured to automatically adjust the shunt, via the at least one shunt drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined pressures.

9. The coreless induction furnace of claim 5, further comprising: at least one load cell operably engaged with the shunt and the at least one shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; at least another load cell operably engaged with the shunt and the at least another shunt drive assembly of each adjustable shunt system of the set of adjustable shunt systems; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one shunt drive assembly, and the at least another shunt drive assembly; wherein the controller is configured to automatically adjust the shunt, via at least one of the at least one shunt drive assembly and the at least another shunt drive assembly, when at least one of the at least one load cell and the at least another load cell measures pressure that is outside a range of predetermined pressures.

10. The coreless induction furnace of claim 1, further comprising: an adjustable head system operably engaged with the outer shell; wherein the adjustable head system is configured to automatically adjust at least a head of the adjustable head system relative to the at least one cooling coil and the power coil; and wherein the set of adjustable shunt systems and the adjustable head system are automatically adjustable independently and separately from one another.

11. The coreless induction furnace of claim 10, wherein the adjustable head system comprises: a head; an apron operably engaged with the outer shell; and at least one head drive assembly operably engaged with the apron; wherein the at least one head drive assembly is configured to automatically adjust the head along an axis relative to the apron.

12. The coreless induction furnace of claim 11, wherein when the head is adjusted by the at least one head drive assembly, the at least one cooling coil and the power coil are adjusted with the head.

13. The coreless induction furnace of claim 11, wherein the at least one head drive assembly comprises: a first jack having a first end that operably engages with the head and a second end opposite to the first end; and a first motor operably engaged with jack; wherein the first motor is configured to linearly move the first jack and the head along the axis relative to the apron.

14. The coreless induction furnace of claim 11, wherein the adjustable head system further comprises: at least another head drive assembly operably engaged with the apron and the head; wherein the at least another head drive assembly is configured to automatically adjust the head along a second axis relative to the apron and parallel with the axis of the at least one head drive assembly; and wherein each of the at least one head drive assembly and the at least another head drive assembly adjusts the head independently and separately from one another.

15. The coreless induction furnace of claim 11, wherein the head of the adjustable head system further comprises: a top plate operably engaged with the at least one head drive assembly; and a refractory member operably engaged with the top plate and configured to engage with the at least one cooling coil to maintain a desired pressure on the at least one cooling coil and the power coil; wherein the top plate and the refractory member are adjustable relative to the apron and the at least one cooling coil.

16. The adjustable shunt system of claim 11, further comprising: at least one load cell operably engaged with the head and the at least one head drive assembly; and a controller electrically connected with the at least one load cell and the at least one head drive assembly; wherein the controller is configured to automatically adjust the head, via the at least one head drive assembly, when the at least one load cell measures a pressure that is outside a range of predetermined

pressures.

17. The adjustable shunt system of claim 14, further comprising: at least one load cell operably engaged with the head and the at least one head drive assembly; at least another load cell operably engaged with the head and the at least another head drive assembly; and a controller electrically connected with the at least one load cell, the at least another load cell, the at least one head drive assembly, and the at least another head drive assembly; wherein the controller is configured to automatically adjust the shunt, via at least one of the at least one head drive assembly and the at least another head drive assembly, when at least one of the at least one load cell and the at least another load cell measure a pressure that is outside a range of predetermined pressures.

18. A method of automatically adjusting one of a shunt of a shunt system and a head of a head system of a coreless induction furnace, comprising steps of: engaging the shunt system with an outer shell of the coreless induction furnace; engaging the head system with the outer shell of the coreless induction furnace; engaging at least one cooling coil of the coreless induction furnace with the shunt system and the head system; engaging a power coil of the coreless induction furnace with the shunt system and the at least one cooling coil; and automatically adjusting one of the shunt of the shunt system and a head of the head system of the coreless induction furnace.

19. The method of claim 18, wherein the steps of engaging the shunt system with the outer shell, the at least one cooling coil, and the power coil further comprises: engaging a support column of the shunt system with the outer shell; engaging at least one shunt drive assembly of the shunt system with the support column of the shunt system; engaging the at least one shunt drive assembly with the shunt of the shunt system; engaging the at least one cooling coil of the coreless induction furnace and the power coil of the coreless induction furnace with the shunt; and automatically adjusting the shunt of the shunt system of the coreless induction furnace, via the at least one shunt drive assembly, relative to the at least one cooling coil and the power coil.

20. The method of claim 18, wherein the steps of engaging the head system with the outer shell and the at least one cooling coil further comprises: engaging an apron of the shunt system with the outer shell; engaging at least one head drive assembly of the head system with the apron of the head system; engaging the at least one head drive assembly with the head of the head system; engaging at least one cooling coil of the coreless induction furnace with the head; and automatically adjusting the head of the head system of the coreless induction furnace, via the at least one head drive assembly, relative to the at least one cooling coil.
